

THIS IS A SYNTHETIC WORLD: THREE ESSAYS ON CAUSAL INFERENCE, APPLIED
ECONOMETRICS, AND MACHINE LEARNING FOR POLICY ANALYSIS

by

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DEDICATION

For my family, friends, and mentors.

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CHAPTER 1

Paradise Lost? Estimating the Economic Impact of Hawaii's Mandatory Quarantine using Synthetic Historical Controls ABSTRACT

I estimate the causal impact of Hawaii's March 2020 mandatory quarantine on its tourism industry and labor market using the Synthetic Historical Control method. Analyzing visitor days, hotel occupancy, accommodations employment, and the Coincident Economic Activity Index (all in annualized year-over-year growth rates), I find substantial negative impacts from the quarantine. Observed outcomes fall significantly outside 90% conformal out-of-sample prediction intervals, indicating a treatment effect. Placebo tests further validate the model's internal credibility. This research demonstrates the method's utility and informs the broader debate on economic impacts of anti-COVID policy.

1.1 Introduction

Among all U.S. states, Hawaii implemented perhaps the most extreme and comprehensive economic shutdown in response to the COVID-19 pandemic. This paper studies the short-run economic consequences of Hawaii's March 2020 mandatory quarantine ([Yan et al., 2022](#)). Unlike most regions that experienced economic contraction from COVID-19 through voluntary distancing, fear, or uneven mandates, Hawaii implemented one of the few deliberate and coordinated state-level shutdowns in the United States. This policy was aimed not merely at mitigating the virus, but at suppressing travel and commerce altogether. On March 26, 2020, the state imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its

borders.

This paper addresses a core policy question: What are the economic costs of decisive early action in the face of an uncertain crisis? While much has been written about the epidemiological and social effects of COVID-19 interventions, less is known about their economic toll. This is especially true in cases where policymakers opted for maximal containment. The most famous case was Wuhan’s January 2020 lockdown, as studied by [Ke and Hsiao \(2022\)](#). However, work outside of this context is rare. Understanding these costs is essential not just for pandemic preparedness, but for future crises, such as climate shocks or geopolitical disruptions, where early intervention may again carry significant short-run economic risks in exchange for long-term societal benefits ([Newman and Noy, 2023](#)).

While jurisdictions like California and New York have been extensively studied, Hawaii, despite enacting arguably more stringent controls, remains curiously absent from empirical analyses. As such, Hawaii offers a rare glimpse into how a near-zero-COVID policy—one that *actually* shut down travel and commerce—might play out within a U.S. federal system. While other states (such as Texas and Georgia) issued stay-at-home orders or imposed limited business closures, few states enforced them strictly, and most reopened early under political pressure. Hawaii, by contrast, implemented a full-scale quarantine and border closure, a level of intervention unmatched by any other U.S. jurisdiction. In that sense, its strategy might be called “The Great Isolation with Aloha characteristics”, even if it was not quite as extreme as Wuhan.

Despite its policy distinctiveness and the strength of its intervention relative to the mainland,

Hawaii has been largely excluded from existing empirical evaluations. This is partly due to the context of the policy, given the lack of suitable control units to draw comparisons from. Many of the most pressing policy questions in public policy and economics concern one-time, systemic shocks: What would the United States' economy have looked like absent the American revolution? What would have happened to unemployment in Great Britain without Brexit? In these cases, credible untreated units often do not exist, making standard causal inference tools, like difference-in-differences (DiD) and synthetic control (SCM), difficult or impossible to apply.

The global nature of COVID-19 presents a core challenge for causal inference: nearly every plausible comparison unit also experienced some degree of shock. In Hawaii's case, its singular geographic and economic features further preclude credible external controls. This motivates the use of Synthetic Historical Control (SHC) method by [Chen et al. \(2024\)](#). SHC avoids the need for external control units by leveraging the treated unit's own pre-treatment dynamics, allowing credible counterfactual construction even when there are no geographic donors to leverage. Empirically, I find that Hawaii experienced substantial—but mostly short-lived—economic losses across the tourism and labor sectors, easing by early 2021.

The remainder of the paper proceeds as follows. Section [1.2](#) reviews prior research on early pandemic interventions and outlines key methodological challenges in studying one-off, jurisdiction-wide policies. Section [1.3](#) presents the SHC framework and its application in this setting. Section [1.4](#) describes the data. Section [1.5](#) reports the empirical results. Section [1.6](#) discusses broader implications for public policy and crisis response.

1.2 Background

1.2.1 Tourism in Hawaii

Hawaii's economy transitioned from a plantation-dominated model in the late 19th and early 20th centuries to tourism as the primary driver after statehood. The sugar and pineapple industries, controlled by a few powerful firms known as the "Big Five," peaked in the 1960s but declined due to global competition, labor costs, and diversification efforts. Statehood in 1959, combined with the introduction of jet air travel, spurred a tourism boom, with visitor numbers surging from 171,000 in 1958 to over 6.7 million by 1990 ([Croix, 2021](#)). This shift modernized the economy, increased population growth, and generated substantial revenue, but it also created dependency on external factors like air travel and international markets.

Tourism is central to Hawaii's fiscal health, generating \$17.75 billion in visitor spending in 2019 alone, which supported \$2.07 billion in state tax revenue and 216,000 jobs across direct, indirect, and induced sectors. On any given day in 2019, there were 249,000 visitors contributing \$48.6 million in spending. Key islands like O'ahu (22.4 million daily spending) and Maui (14.0 million) bear the brunt of this activity ([Hawai'i Tourism Authority, 2019](#)). Leisure and hospitality employment, a proxy for tourism-related jobs, hovered around 190,000-200,000 annually pre-2020, representing about one-third of the state's workforce. This reliance amplifies risks from shocks, as seen in past recessions, but also underscores tourism's role in funding public services and infrastructure.

Hawaii implemented some of the earliest and most aggressive COVID-19 containment policies

in the United States. On March 4, 2020, Governor David Ige issued an emergency proclamation officially recognizing the virus as a statewide threat ([Maui Now Staff, 2020](#)). Within weeks, the state began taking unprecedented measures alongside states like California ([Friedson et al., 2021](#)). On March 17, Governor Ige publicly urged all visitors to suspend their trips to Hawaii for 30 days, anticipating substantial economic damage to the state’s tourism-dependent economy. *"We do know there will be significant impact,"* he stated, though no concrete estimates were available at the time ([Nakaso, 2020](#)). A sweeping stay-at-home order followed on March 25, banning non-essential activities, closing non-critical businesses, and effectively shutting down day-to-day life and travel across the islands ([HNN Staff, 2020](#)). These actions were implemented alongside a mandatory 14-day quarantine on all arrivals, passed the same week, which further intensified the shutdown and isolated the state from both domestic and international travel. Despite the unprecedented scope of these interventions, no formal quantitative analysis has yet estimated the short-run economic effects of Hawaii’s approach. This paper provides such an assessment.

1.2.2 Literature Review

The onset of the COVID-19 pandemic spurred a global wave of NPIs, with governments acting swiftly to limit viral transmission. Much of the early research focused on the public health rationale behind these policies—particularly the value of acting early to prevent exponential growth in cases. [Friedson et al. \(2021\)](#), for instance, study California’s shelter-in-place order and argue that its primary purpose was to delay the pandemic’s peak, buying time for hospitals to prepare. They emphasize the importance of California’s timing—implementing

the policy while infection growth was still modest—as a key to its effectiveness.

Other studies reinforce the value of early, aggressive containment. Using SCM, [Yang \(2021\)](#) analyzes Wuhan’s lockdown and estimate substantial reductions in population movement and COVID-19 transmission, emphasizing the relevance of their findings for international public health policy. [Cerqueti et al. \(2021\)](#), studying Italy’s decision to close schools, stress that timing is critical when dealing with exponential contagion. They also note a political economy challenge: if containment is successful, the intervention may appear excessive in retrospect, raising political costs for early action. [Bayat et al. \(2020\)](#) offer a quantitative example, estimating that deaths in New York could have been reduced by over 80% had lockdown been enacted ten days earlier. Outside of China and Europe, similar insights emerge. [Sobiech Pellegrini \(2022\)](#) and [Li and Shankar \(2023\)](#) document similar patterns in Brazil and Texas, finding that premature reopenings by state governments led to sustained spikes in cases.

Collectively, these studies underscore a key lesson: early interventions can mitigate public health disasters, but often at economic cost. This tradeoff between health and the economy became central to both policymaking and public discourse. [Osman et al. \(2022\)](#) document how opponents of lockdowns, particularly in conservative circles, criticized them as economically inefficient and ineffective. [Gibson and Sun \(2023\)](#) evaluate statewide stay-at-home orders using synthetic control, focusing on new jobless claims as an economic indicator of interest. [Lee and Yang \(2022\)](#) examine the labor market impacts of COVID-19 in South Korea, highlighting the economic consequences of NPIs. [Ke and Hsiao \(2022\)](#) study the economic

impact of Hubei’s lockdown. They show that while Hubei’s lockdown caused a sharp downturn in the short term, the effects were short-lived—suggesting that delay or inaction might have led to more lasting harm.

Other work has assessed broader welfare outcomes of interventions. [Cole et al. \(2020\)](#) show how the Wuhan lockdown reduced air pollution and associated mortality using augmented SCM. [Oliu-Barton et al. \(2022\)](#) study vaccine mandates in Europe and estimate that increased vaccine uptake not only averted deaths but also prevented GDP losses—suggesting that public health policy can have broader economic upside when well-designed. Despite the breadth of this literature, Hawaii’s case remains notably absent. Yet it represents perhaps the clearest U.S. example of an early, and most aggressive intervention. This paper contributes by bringing Hawaii into the empirical record and by offering credible estimates of the short-run economic costs of a “zero-COVID” style border ([Yuan, 2022](#)).

While much has been learned about the effects of pandemic interventions, measuring their economic impacts remains challenging. Most studies evaluating anti-COVID policies rely on difference-in-differences (DiD) or synthetic control methods (SCM). Where cross-sectional variation in policy exists—e.g., some counties or states implementing lockdowns while others do not—SCM and panel methods have been used productively (e.g., [Yang, 2021](#); [Ke and Hsiao, 2023](#); [Gong et al., 2024](#)). However, these designs depend on the availability of unaffected or minimally affected units to serve as comparisons. As noted by [Callaway and Li \(2023\)](#) and [Bilgel \(2022\)](#), this and other critical assumptions often break down in the context of a global pandemic, where nearly all regions experienced some form of shock.

These limitations apply especially in Hawaii’s case. Every other U.S. state (as well as comparable island economies) faced their own pandemic-related disruptions. There is no unaffected or even roughly comparable unit that can serve as a credible counterfactual for Hawaii’s tourism and labor markets. In such cases, standard cross-sectional methods fail. To my knowledge, the only study examining Hawaii’s economic response to the pandemic is [Bond-Smith and Fuleky \(2022\)](#), which provides valuable descriptive analysis but does not construct counterfactual estimates.

To address this challenge, I apply the SHC method developed by [Chen et al. \(2024\)](#). SHC circumvents the need for external control units by leveraging the treated unit’s own pre-intervention history. Rather than borrowing information from across space, it borrows strength across time, enabling credible inference in settings where conventional comparative approaches are infeasible—precisely the situation posed by Hawaii’s quarantine.

1.3 Methodology

We know what happened given that Hawaii mandated two-week quarantines. We do not know what would have happened had they not done this. Given its novelty and radical deviation from SCM, this section summarizes the SHC method by [Chen et al. \(2024\)](#). For a comprehensive treatment of the method, including the theoretical proofs and asymptotic properties, readers are referred to [Chen et al. \(2024\)](#).

Notation Let $\mathbb{R}$ denote the set of real numbers, and let $\|\cdot\|_1$ denote the usual ℓ_1 vector norm. Throughout, I denote sets using calligraphic letters (e.g., $\mathcal{T}, \mathcal{S}, \mathcal{N}$) for clarity. Scalars are

represented using plain lowercase letters (e.g., h, t, n, m), and matrices are denoted by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{W}, \mathbf{Y}$). Given a vector $\mathbf{x} = (x_1, x_2, \dots, x_T)^\top \in \mathbb{R}^T$ and an index set $\mathcal{I} \subseteq \{1, \dots, T\}$, I write $\mathbf{x}_{\mathcal{I}} := (x_t)_{t \in \mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$ to denote the subvector of $\mathbf{x}$ corresponding to indices in $\mathcal{I}$, preserving their original order.

Let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. To reconstruct the evaluation window, we compare it against earlier segments of the treated unit's own pre-treatment trajectory.

Define the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$ as a collection of N overlapping length- m subvectors extracted from the smoothed pre-treatment series. Each column is a contiguous subvector of the form $\widetilde{\mathbf{y}}_{[i, i+m-1]}$, with i ranging over eligible start points in $\mathcal{T}_1$ that precede the evaluation window. The precise construction is given in [Section 1.3.2](#).

1.3.1 Synthetic Historical Control Method

Potential Outcomes Following the Rubin potential outcomes framework, let y_t^1 and y_t^0 denote the potential outcomes for the treated unit at time $t \in \mathcal{T}$, corresponding to the outcomes under treatment and no treatment, respectively. The observed outcome at time t is $y_t = d_t y_t^1 + (1 - d_t) y_t^0$, where $d_t = \mathbf{1}\{t > T_0\}$. The fundamental problem of causal inference is that for $t > T_0$, we observe only the treated potential outcome $y_t^1 = y_t$, while the untreated potential outcome y_t^0 remains unobserved. Let the SHC estimator for y_t^0 be denoted $\hat{y}_t^0 := y_t^{\text{SHC}}$ for all $t \in \mathcal{T}_2$. Define the observed post-treatment outcome vector as $\mathbf{y}_{\text{post}} := (y_t^1)_{t \in \mathcal{T}_2}$ and the estimated counterfactual vector as $\mathbf{y}_{\text{post}}^0 := (\hat{y}_t^0)_{t \in \mathcal{T}_2}$. The out-of-sample treatment effect is

$$\alpha_t := y_t^1 - y_t^0, \quad \text{for } t \in \mathcal{T}_2, \quad (1.1)$$

with estimator $\hat{\alpha}_t := y_t - \hat{y}_t^0$. Stacking over $t \in \mathcal{T}_2$ yields:

$$\hat{\boldsymbol{\alpha}} := \mathbf{y}_{\text{post}} - \mathbf{y}_{\text{post}}^0. \quad (1.2)$$

The average treatment effect on the treated (ATT) is then given by:

$$\widehat{\text{ATT}} := \frac{1}{n} \sum_{t=T_0+1}^{T_0+n} (y_t - \hat{y}_t^0) = \frac{1}{n} \mathbf{1}^\top \hat{\boldsymbol{\alpha}}, \quad (1.3)$$

where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.

1.3.2 The Model

The SHC method introduced by [Chen et al. \(2024\)](#) constructs a counterfactual using the treated unit's own pre-treatment time series.¹ SHC requires a few assumptions: [Assumption 1](#), [Assumption 2](#), and [Assumption 3](#).

Assumption 1 (Error Properties). *The error term satisfies $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t^2] < \infty$ for all $t \in \mathcal{T}$.*

[Assumption 1](#) says unmodeled fluctuations or noise in the outcome are random.

Assumption 2 (Latent Component Smoothness and Matching Condition). *The latent trend $\tilde{y}(t)$ is $(H + 1)$ times differentiable for some integer $H \geq 3$, with derivative bounded as $|\tilde{y}^{(H+1)}(t)| < \varepsilon$ uniformly over $t \in \mathcal{T}_1$. Additionally, $\exists \mathbf{w}^* \in \Delta^{N-1}$ such that $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre} \mathbf{w}^*$.*

[Assumption 2](#) allows us to approximate the post-treatment latent component using a local linear extrapolation of pre-treatment values and posits that perfect match exists within the donor pool for the evaluation window, much as [Abadie et al. \(2010\)](#) assume about donor pool units matching to the treated unit.

Assumption 3 (Feasibility). *There exists a weight vector $\mathbf{w} \in \Delta^{N-1}$ such that $\hat{\mathbf{y}}_{eval} = \widetilde{\mathbf{Y}}_{pre} \mathbf{w}$, with $\|\mathbf{w}\|_0 \leq s \ll N$. Let $\widetilde{\mathbf{Y}}_{pre}^{(s)}$ denote the submatrix formed by the active columns of $\widetilde{\mathbf{Y}}_{pre}$ indexed by $\text{supp}(\mathbf{w})$; then its Gram matrix satisfies $(\widetilde{\mathbf{Y}}_{pre}^{(s)})^\top \widetilde{\mathbf{Y}}_{pre}^{(s)} \succ 0$.*

¹SCM estimates the counterfactual by solving:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\text{argmin}} \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2, \quad \forall t \in \mathcal{T}_1, \quad (1.4)$$

where $\mathbf{Y}_0 \in \mathbb{R}^{T_0 \times N_0}$ is the donor matrix of unexposed units and $\Delta^{N_0} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} : \|\mathbf{w}\|_1 = 1 \right\}$ is the probability simplex.

Chen et al. (2024) note that [Assumption 3](#) is the sample-variant of [Assumption 2](#). It says that the SHC weights are identifiable, sparse, and numerically stable, similar to the assumptions described by [Abadie \(2021\)](#).

The first step of the SHC method smooths $\mathbf{y}_{\text{pre}}$ using local linear regression, where we fit the regression model:

$$y_j \approx a_t + b_t(j - t), \quad \text{for } j = 1, \dots, T_0, \quad (1.5)$$

with local intercept $a_t \in \mathbb{R}$ and slope $b_t \in \mathbb{R}$. Define the centered time vector $\mathbf{s}_t = (1 - t, 2 - t, \dots, T_0 - t)^\top \in \mathbb{R}^{T_0}$, and the design matrix $\mathbf{X}_t = [\mathbf{1} \ \mathbf{s}_t] \in \mathbb{R}^{T_0 \times 2}$. The diagonal kernel weight matrix is:

$$\mathbf{W}_t = \text{diag}(w_{1t}, w_{2t}, \dots, w_{T_0t}), \quad w_{jt} \propto \exp\left(-\frac{1}{2} \left(\frac{j - t}{h}\right)^2\right), \quad (1.6)$$

where $(h > 0)$ is a bandwidth parameter governing smoothness. The local coefficients $\boldsymbol{\beta}_t = (a_t, b_t)^\top$ solve quadratic programming problem that is strictly convex

$$\boldsymbol{\beta}_t = \underset{\boldsymbol{\beta} \in \mathbb{R}^2}{\text{argmin}} \left\| \mathbf{W}_t^{1/2} (\mathbf{y}_{\text{pre}} - \mathbf{X}_t \boldsymbol{\beta}) \right\|_2^2. \quad (1.7)$$

Fortunately, this optimization admits a closed-form solution for the coefficients:

$$\boldsymbol{\beta}_t = \left(\mathbf{X}_t^\top \mathbf{W}_t \mathbf{X}_t \right)^{-1} \mathbf{X}_t^\top \mathbf{W}_t \mathbf{y}_{\text{pre}}. \quad (1.8)$$

The smoothed outcome at time t is $\tilde{y}_t = a_t = \mathbf{e}_1^\top \boldsymbol{\beta}_t$, where $\mathbf{e}_1 = (1, 0)^\top$. Let $\tilde{\mathbf{y}} =$

$(\tilde{y}_1, \dots, \tilde{y}_{T_0})^\top \in \mathbb{R}^{T_0}$ denote the full smoothed series. After we have smoothed over the pre-treatment period before the evaluation period, we build our donor matrix of historical control segments/units.

In the second step, SHC constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m . Each segment $\tilde{\mathbf{y}}_{[i, i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix $\tilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \tilde{\mathbf{y}}_{[1, m]} & \tilde{\mathbf{y}}_{[2, m+1]} & \dots & \tilde{\mathbf{y}}_{[N, N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}$.

Example 1 (Constructing $\tilde{\mathbf{Y}}_{\text{pre}}$). *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months in order to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is: $N = T_0 - m - n + 1 = 60 - 12 - 6 + 1 = 43$. Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.*

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\text{argmin}} \quad & \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1 \end{aligned} \tag{1.9}$$

[Chen et al. \(2024\)](#) note that in the general case when we have more donor units than we do the length of the segments ($N > m$), the optimization becomes an ill-posed problem. The regularization term stabilizes the optimization by penalizing weight allocations along directions of low variance in the donor pool, reducing sensitivity to collinear segments. Formally, $\mathbf{C}_0$ contains the eigenvectors of the Grammian matrix $\mathbf{G} \in \mathbb{R}^{N \times N}$. $\mathbf{C}_0$ corresponds

to the eigenvalues less than $\tau > 0$. These directions capture low-variance combinations of donor segments, and the penalty shrinks weights in those directions.² The corresponding weight vector $\mathbf{w}^*$ is used to compute the counterfactual by applying it to the shifted donor segments:

$$\widetilde{\mathbf{Y}}_{\text{post}} = \begin{bmatrix} \widetilde{\mathbf{y}}_{[1+m, m+n]} & \widetilde{\mathbf{y}}_{[2+m, m+n+1]} & \cdots & \widetilde{\mathbf{y}}_{[N+m, N+m+n-1]} \end{bmatrix} \in \mathbb{R}^{n \times N}. \quad (1.10)$$

This yields in-sample and out-of-sample predictions for the SHC estimator. The full counterfactual vector is then: $\mathbf{y}^0 = \widetilde{\mathbf{Y}}\mathbf{w}^* \in \mathbb{R}^{m+n}$.

Bandwidth Selection via Cross-Validation To select the bandwidth parameter h used in local linear smoothing of $\mathbf{y}_{\text{pre}}$, I implement a *leave-one-out cross-validation* (LOOCV) procedure over a grid of candidate values $h \in [h_{\min}, h_{\max}] \subset \mathbb{R}_{>0}$. This procedure chooses the bandwidth that minimizes out-of-sample prediction error for the smoothed pre-treatment series $\widetilde{\mathbf{y}} \in \mathbb{R}^{T_0}$, as defined in Equation~(1.8).

For each candidate h , and for each time point $t \in \mathcal{T}_1$, define Gaussian kernel weights

$$w_{\tau t}^{(h)} = \exp\left(-\frac{1}{2}\left(\frac{\tau - t}{h}\right)^2\right), \quad \tau \in \mathcal{T}_1 \setminus \{t\}, \quad (1.11)$$

used to construct the diagonal matrix $\mathbf{W}_t^{(-t)} \in \mathbb{R}^{(T_0-1) \times (T_0-1)}$, omitting the index t . Let $\mathbf{X}_t^{(-t)} \in \mathbb{R}^{(T_0-1) \times 2}$ and $\mathbf{y}_{\text{pre}}^{(-t)} \in \mathbb{R}^{T_0-1}$ denote the design matrix and response vector with time

²The ς (sigma) term is a small positive constant used to stabilize the solution in cases of near multicollinearity.

t excluded.

Then the local linear coefficients $\beta_t^{(-t)} = (a_t^{(-t)}, b_t^{(-t)})^\top$ are obtained by solving Equation~(1.7).

The leave-one-out smoothed value at time t is then:

$$\tilde{y}_t^{(-t)} = a_t^{(-t)} = \mathbf{e}_1^\top \beta_t^{(-t)}, \quad \text{where } \mathbf{e}_1 = (1, 0)^\top. \quad (1.12)$$

To pick the ideal bandwidth, we minimize:

$$h^* = \underset{h \in [h_{\min}, h_{\max}]}{\operatorname{argmin}} \left\{ \operatorname{CV}(h) := \frac{1}{T_0} \sum_{t=1}^{T_0} (y_t - \tilde{y}_t^{(-t)})^2 \right\}. \quad (1.13)$$

Once selected, the optimal bandwidth h^* is used to construct the full smoothed pre-treatment series.

Control Group Selection [Chen et al. \(2024\)](#) note that we should have a parsimonious approach to which historical controls we use, rather than choosing all of them. In their examples, SHC typically requires less than 28 controls. They advocate for a forward selection procedure to choose the optimal control group. I implement a slightly modified variant of this method to choose the ideal number of historical controls. Let $\mathcal{N} = \{1, 2, \dots, N\}$ index the full set of donor segments in the pre-treatment matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. At each step the algorithm selects the donor index $i_j \in \mathcal{N} \setminus \mathcal{S}_{j-1}$ that, when added to the current donor set $\mathcal{S}_{j-1}$, minimizes the in-sample mean squared error (MSE) between the evaluation window

and in-sample fit:

$$\mathcal{S}_j = \mathcal{S}_{j-1} \cup \left\{ \underset{i \in \mathcal{N} \setminus \mathcal{S}_{j-1}}{\operatorname{argmin}} \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S}_{j-1} \cup \{i\})} \mathbf{w}^{(j)} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\}, \quad \mathcal{S}_0 = \emptyset, \quad (1.14)$$

where $\tilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S})}$ denotes the submatrix of $\tilde{\mathbf{Y}}_{\text{pre}}$ with columns indexed by $\mathcal{S}$, and the weights $\mathbf{w}^{(j)} \in \Delta^j$ are the optimal simplex-constrained weights for the selected donor set $\mathcal{S}_j$. This procedure greedily expands the donor set one segment at a time, always choosing the segment that yields the greatest marginal reduction in in-sample error.

To avoid overfitting, [Chen et al. \(2024\)](#) manually choose the number of donors the forward selection procedure should choose. In their case, they experiment with the single best historical segment (which rarely does well), the best 15, 20, and 27 historical control units. However, this is a subjective decision that will yield different results depending on what we select. To mitigate researcher degrees of freedom, I view the number of historical segments as a tuning parameter. I employ a forward selection-based algorithm to choose the number of controls. For my implementation of SHC, I employ an early stopping rule based on a modified Bayesian information criterion (BIC), borrowing inspiration from [Shi and Huang \(2023\)](#). For each donor set size j , I compute:

$$\text{BIC}(j) = m \cdot \log(\text{MSE}_j) + \lambda j, \quad (1.15)$$

where MSE_j is the in-sample error using donor set $\mathcal{S}_j$, and $\lambda = \log(m)$ penalizes the inclusion

of too many donors. The selection procedure terminates when $\text{BIC}(j)$ increases for two consecutive steps. The final donor set $\mathcal{S}^*$ is then taken to be $\mathcal{S}_{j^*}$, where:

$$j^* = \underset{j}{\operatorname{argmin}} \text{BIC}(j). \quad (1.16)$$

This approach adaptively selects a sparse donor subset that closely matches the treated unit's pre-intervention trajectory. The selected donor set $\mathcal{S}^*$ is used to compute the SHC model predictions. I would like to mention that this forward selection method holds much in common with recently proposed methods from the literature. For example, [Shi and Huang \(2023\)](#) propose the forward selected panel data approach which selects the comparison group for the panel data approach by [Hsiao et al. \(2012\)](#) using an aggressive forward selection procedure. [Li \(2024\)](#), correcting for overfitting in the former, proposed the forward Difference-in-Differences estimator, where a forward selection approach is used to select the donor pool for the DiD model. Similar advances have been made SCM, as developed by [Cerulli \(2024\)](#), where a similar selection mechanism is used to estimate the treatment effect. It would be interesting, but is far outside the scope of this work, to compare these estimators to one another and see when their in-sample fits and point estimates agree with one another.

Inference Quantifying uncertainty around the counterfactual predictions is essential for valid inference. [Chen et al. \(2024\)](#) do not provide any inferential method to use aside from in-time placebos. Following recent work in post-modeling inference by [Cattaneo et al. \(2025\)](#), I construct agnostic conformal prediction intervals around the counterfactual series.

These intervals are distribution-free, require no parametric assumptions, and are valid under exchangeability of residuals between the pre- and post-treatment periods (or, good in-sample fit).

Let $\mathbf{y}_{\text{eval}}^0 \in \mathbb{R}^m$ denote the SHC-fitted values during the evaluation period and define the residuals as:

$$\boldsymbol{\varepsilon}_{\text{eval}} := \mathbf{y}_{\text{eval}} - \mathbf{y}_{\text{eval}}^0. \quad (1.17)$$

I then compute the sample mean $\bar{\varepsilon}$ and variance $\hat{\sigma}^2$ of these residuals:

$$\bar{\varepsilon} = m^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} \varepsilon_t, \quad \hat{\sigma}^2 = (m-1)^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} (\varepsilon_t - \bar{\varepsilon})^2. \quad (1.18)$$

To construct prediction intervals for the post-treatment counterfactual $\mathbf{y}_{\text{post}}^0 \in \mathbb{R}^n$, I assume the residuals are sub-Gaussian and apply a conformal bound based on Hoeffding-type concentration inequalities. For a miscoverage rate $\alpha \in (0, 1)$, the interval half-width is given by:

$$c_\alpha = \sqrt{2\hat{\sigma}^2 \log\left(\frac{2}{\alpha}\right)}. \quad (1.19)$$

Then, the agnostic $(1 - \alpha)$ prediction interval for each $t \in \mathcal{T}_2$ is given by:

$$\left[\hat{y}_t^0 + \bar{\varepsilon} - c_\alpha, \hat{y}_t^0 + \bar{\varepsilon} + c_\alpha \right]. \quad (1.20)$$

Stacking these yields lower and upper bound vectors:

$$\mathbf{L} := \mathbf{y}_{\text{post}}^0 + \bar{\varepsilon} \cdot \mathbf{1} - c_\alpha \cdot \mathbf{1}, \quad \mathbf{U} := \mathbf{y}_{\text{post}}^0 + \bar{\varepsilon} \cdot \mathbf{1} + c_\alpha \cdot \mathbf{1}, \quad (1.21)$$

which define the conformal confidence band around the counterfactual trajectory.

This approach allows finite-sample valid coverage under the assumption that the pre- and post-treatment residuals are exchangeable. Intuitively, if the SHC model fits the pre-treatment series well (i.e., residuals are small and centered), then its extrapolations into the post-treatment period are expected to carry similarly bounded uncertainty. In this sense, the residual distribution learned from the evaluation window is used to calibrate uncertainty for the out-of-sample outcomes.

These prediction intervals reflect uncertainty in the estimated counterfactual. Hence, if the observed outcome falls far outside the prediction interval, this is interpreted as evidence of a treatment effect. Unlike traditional confidence intervals around treatment effect estimates (e.g., using standard errors), the conformal intervals here do not assume a data-generating model. I use a default coverage level of 90% (i.e., $\alpha = 0.1$) in all applications.

1.4 Data

To evaluate the impact of Hawaii’s COVID-19 border closure on its tourism economy, I use monthly time series spanning multiple economic dimensions, focusing on four primary outcomes: two related to tourism demand and two capturing broader economic impacts.

All outcomes are transformed into annualized year-over-year (YoY) growth rates. This transformation mitigates nonstationary trends—crucial for the validity of the SHC estimator (Chen et al., 2024)—and allows treatment effects to be interpreted directly as percentage point changes.

The first two outcomes capture the demand side of Hawaii’s tourism sector. Visitor Days measures the total number of person-days spent by visitors in a given month, integrating both the volume of arrivals and their length of stay. Occupancy refers to the statewide hotel occupancy rate, defined as the percentage of available hotel rooms that are occupied. Together, these indicators provide complementary measures of tourism demand: the former captures total visitor presence, while the latter reflects realized capacity utilization across the hospitality sector. Both series are sourced from Hawaii’s Department of Business, Economic Development & Tourism (DBEDT) and span from January 1990 to December 2020.

The remaining outcomes capture economic effects, one sector-specific and one broader in scope. *Accommodation Employment* tracks the number of employees in Hawaii’s accommodation industry, a labor-intensive sector directly tied to tourism activity. *The Coincident Economic Activity Index* summarizes statewide macroeconomic conditions based on four indicators: nonfarm payroll employment, the unemployment rate, average weekly hours worked in manufacturing, and wage and salary disbursements. This index is benchmarked to match the trend of gross state product, making it a comprehensive indicator of the state’s economic health. Both economic indicators are sourced from the Federal Reserve Economic Data (FRED) system and cover January 1991 to December 2020.

Hawaii’s mandatory quarantine policy took effect in late March 2020. I define the intervention to begin in March 2020, corresponding to time $t = 362$, and formalize the treatment indicator as $d_t := \mathbf{1}\{t \geq 362\}$, so that $d_t = 1$ for all months after the policy’s implementation and $d_t = 0$ otherwise. The final pre-treatment period is February 2020, yielding a pre-treatment sample of $T_0 = 361$ months.

1.5 Results

In Figure 1.1 we visualize the outcomes of interest over time from January of 1991 to December 2020. We can see that the growth rate trends for all outcomes were relatively smooth across the full pre-intervention series, exhibiting seasonality and relatively low variance. We also see that the outcomes of interest precipitously dropped when the mandatory quarantine policy was implemented. The plot also suggests that [Assumption 1](#) may be valid. For example, for [Assumption 1](#) to be violated, we would need a situation where the outcome had extremely high variance. While we see seasonality across all outcomes, the time series seemed smooth enough to be approximated by pre-treatment observations.

Figure 1.2 presents the initial SHC results. For these results, $m = 60$ (5 years of evaluation data), but the main results remain robust up until 8 years of evaluation data, with slight deviations in uncertainty.

Across all four primary outcomes, I find large and precisely estimated negative effects from Hawaii’s border closure. Note that the good in-sample fit suggests that [Assumption 2](#) is satisfied. Tourism demand was the most directly affected: Visitor Days declined by 85.4

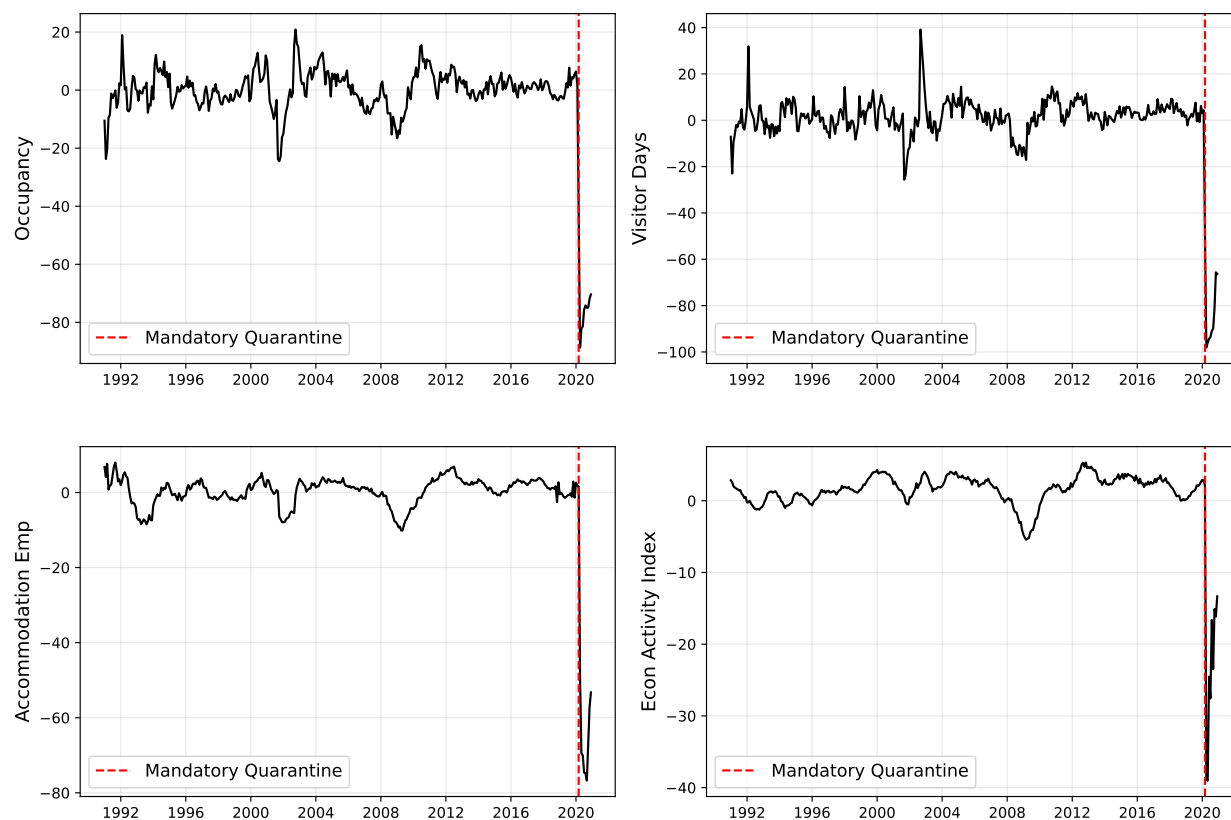


Figure 1.1: Outcome Trends.

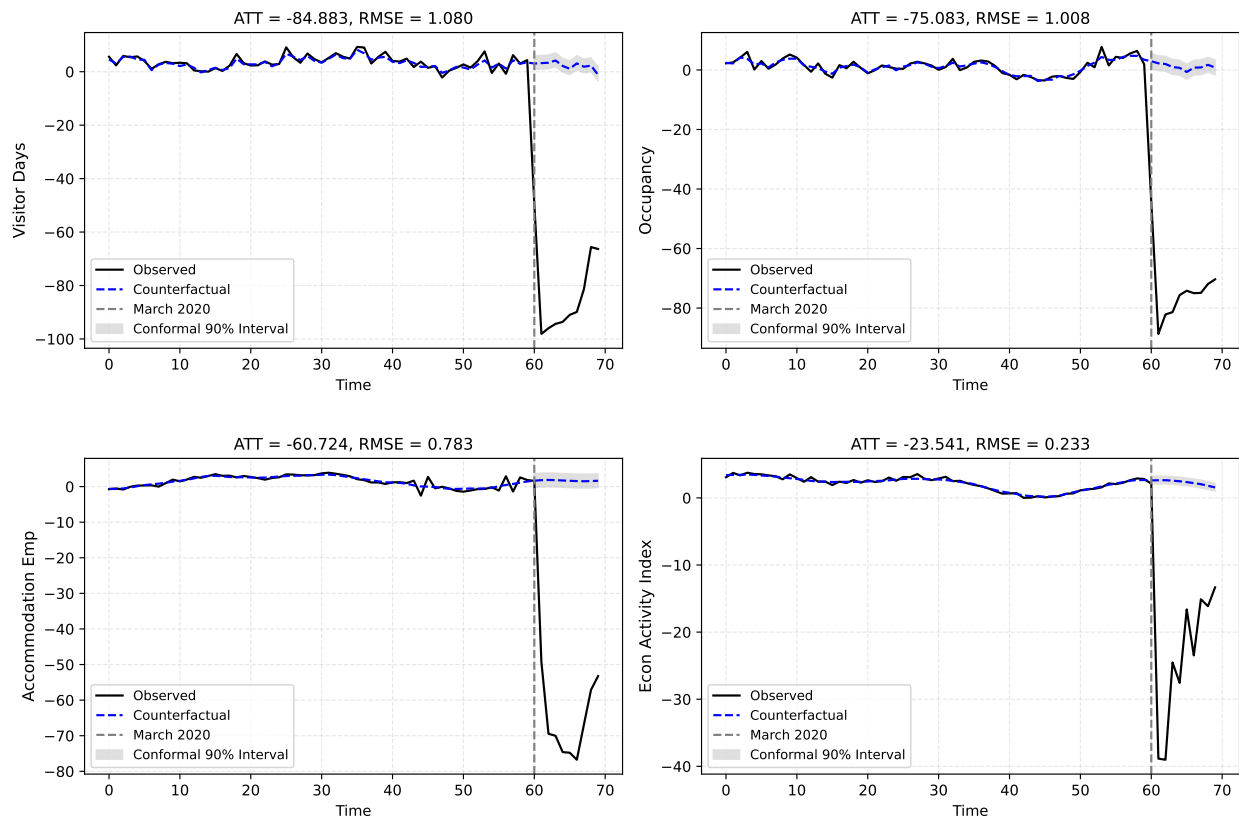


Figure 1.2: SHC Analysis.

percentage points, and Hotel Occupancy fell by 75.8 percentage points, relative to their synthetic historical counterfactuals. These are dramatic, near-total collapses in the state’s primary economic driver. The SHC model achieves a close pre-treatment fit for both outcomes (RMSEs ≈ 1), suggesting that the model is well-calibrated to detect meaningful post-treatment divergence.

Labor market and broader economic indicators reflect these shocks. Accommodation employment declined by 60.7 percentage points, while the Economic Activity Index fell by 27.7 percentage points. Though these outcomes are more aggregated, they follow the same pattern as the tourism indicators—abrupt declines followed by a gradual recovery towards the end of 2020, suggesting Hawaii’s shutdown sent a measurable shock through the broader economy. Notably, in all four cases, the observed values after March 2020 fall well outside the 90% conformal prediction intervals, strengthening the inference that these declines reflect true treatment effects rather than model noise or volatility.

The results also suggest the mechanics of the causal effect: the sharp decline in overall employment, especially in accommodations, is plausibly driven by the collapse in tourism. Given Hawaii’s economic structure, the vast majority of hotel/hospitality employment is directly demand-driven. Thus, the drop in visitor-days and occupancy rates leads to reduced staffing needs across the sector. This channel provides a natural explanation for the labor market contraction observed in the data. Importantly, the pattern across all outcomes suggests a consistent causal chain: border closure leads to a collapse in tourism, which in turn leads to employment contraction, macroeconomic slowdown, and a fiscal shock. This

sequential relationship reinforces the credibility of the estimates and gives confidence that the economic effects documented here reflect real-world consequences of the policy intervention. Robustness checks support this interpretation. Using 10 years of pre-treatment data produces nearly identical ATT estimates, with only minor degradation in early-period fit. The central pattern—a sharp short-run shock followed by recovery—holds across specifications. The results are presented in Figure 1.3.

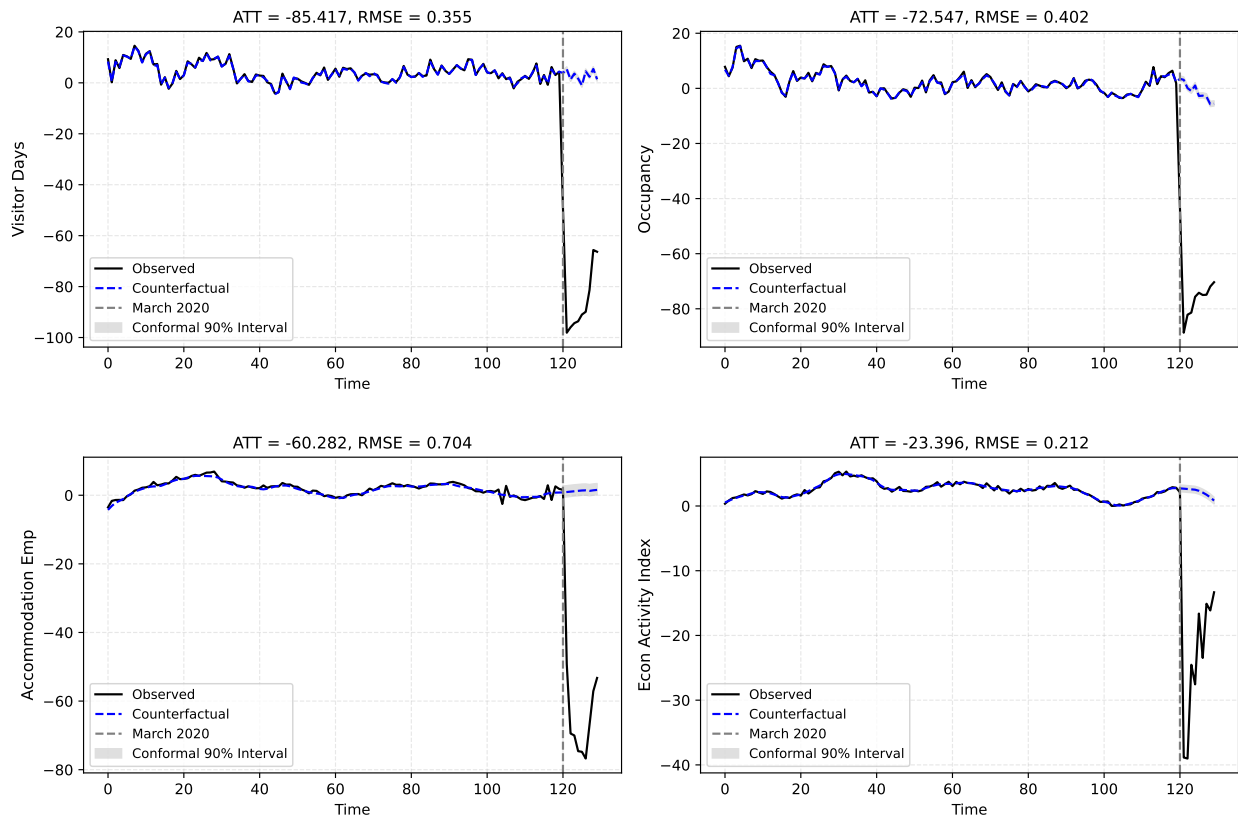


Figure 1.3: SHC Analysis, $m = 120$.

Placebo tests using a September 2019 “fake” intervention date yield negligible effects, with estimated ATTs well within the model’s baseline error margins. This provides reassurance that the model is not spuriously detecting treatment effects where none exist. Figure 1.4

presents these results.

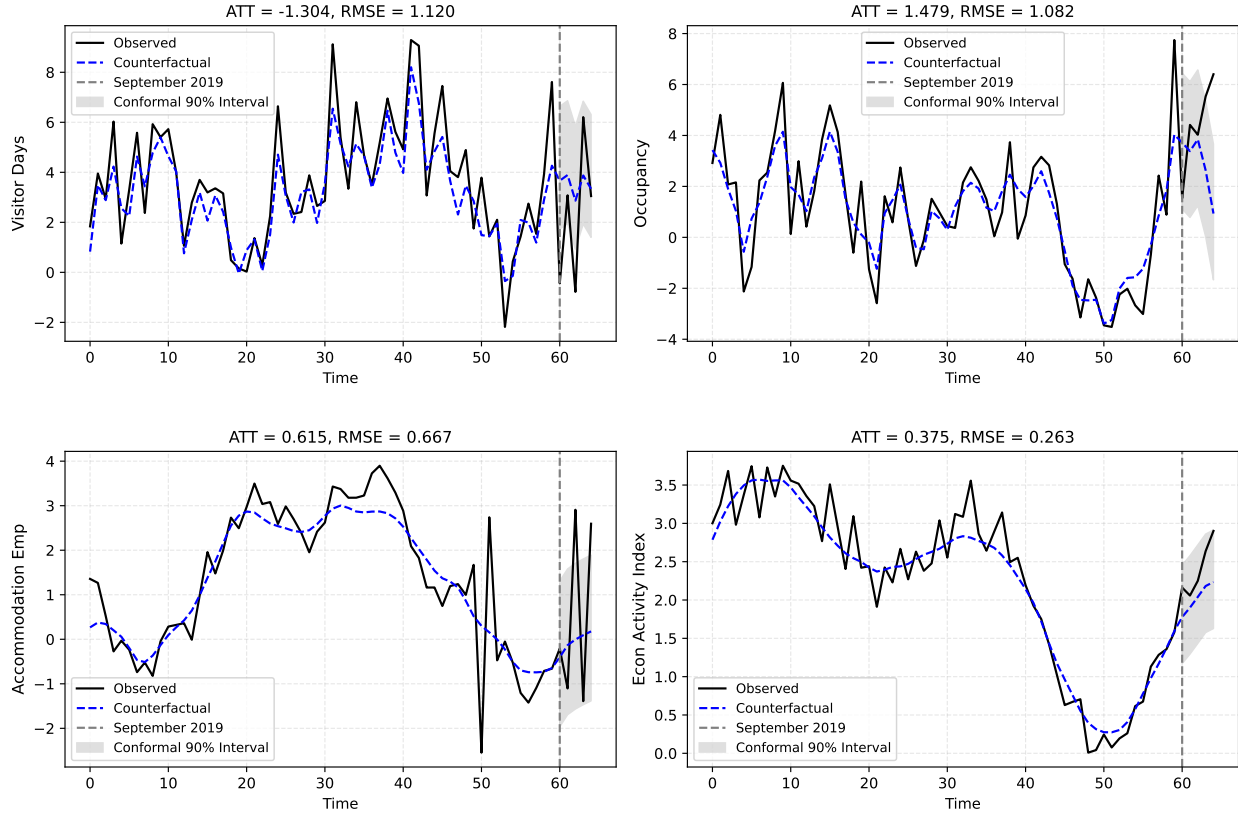


Figure 1.4: Placebo Studies.

These results offer two key takeaways. First, the SHC model does not erroneously detect treatment effects when none exist—it exhibits strong temporal specificity. Second, even in cases where placebo ATTs are slightly nonzero, they fall within normal forecast uncertainty. This is confirmed by the conformal prediction intervals, which entirely cover the observed outcomes during the placebo window. That is, any apparent deviation between predicted and realized values is consistent with random variation under no treatment. These placebo checks strengthen the case for a causal interpretation of the post-March 2020 findings: the observed deviations are not artifacts of overfitting or model instability, but genuine responses

to Hawaii's border closure.

To estimate the impact of the mandatory quarantine on fiscal losses, I collect data on taxes generated from Hawaii's General Excise tax for the hotel industry and overall transient accommodations, using quarterly revenue data from 2010 to 2019 (with the pre-treatment period being from 1991 to the end of 2009). I estimate a 91.7 percentage point decline in Transient Accommodations Tax receipts, and an 88.3 point drop in hotel-related General Excise Taxes, consistent with large-scale fiscal losses during the quarantine period. These results are presented in Figure 1.5.

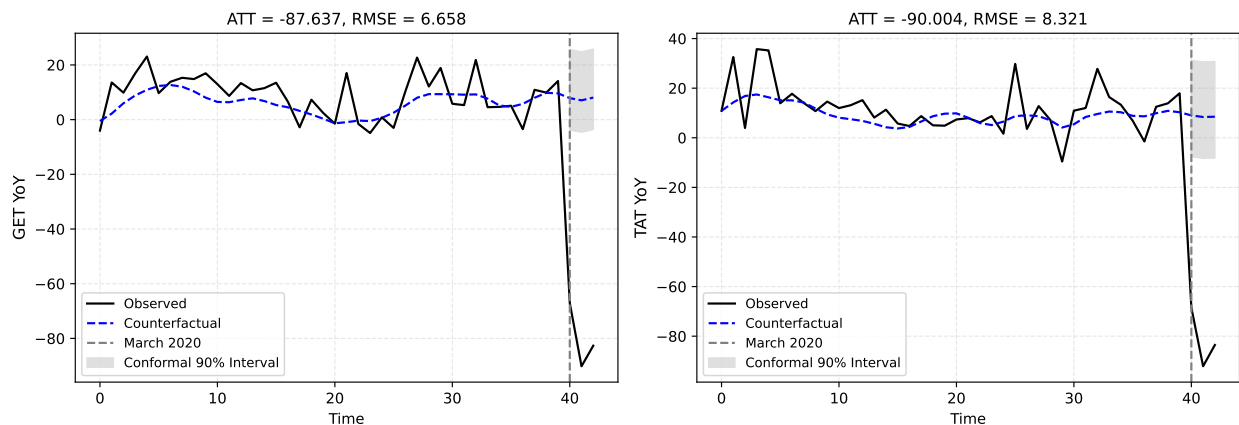


Figure 1.5: Additional Outcomes.

1.6 Discussion

This paper provides a quantitative estimate of the short-run economic costs of an early, aggressive pandemic response. Hawaii's mandatory quarantine policy caused a near-total collapse in tourism, with sharp but largely temporary contractions in employment and broader economic activity. Despite its severity, these effects were not persistent: most metrics

rebounded by early 2021. This pattern—steep initial losses followed by gradual recovery—suggests that decisive early action may impose short-run pain but help avoid longer-term disruption.

These results contribute to an ongoing debate over whether early containment policies are worth their economic cost. Hawaii’s case supports the theory of “front-loaded sacrifice”: by absorbing short-term damage, the state avoided sustained epidemiological and economic chaos. This interpretation echoes findings from [Ke and Hsiao \(2022\)](#) on Hubei’s lockdown and aligns with the intuition of models emphasizing early interventions to delay or avoid exponential harm.

[Bond-Smith and Fuleky \(2022\)](#) provide useful context. They document that, as of 2022, Hawaii’s non-farm employment remained nearly 10% below pre-pandemic levels, raising concerns about persistent scarring. Yet they also note that Hawaii avoided major COVID waves until Delta and Omicron. This contrast—between early economic contraction and delayed epidemiological impact—underscores the tradeoff central to this paper: early action can buy time, but at an economic cost.

This study does not estimate public health benefits directly, nor does it claim Hawaii’s policy was cost-effective overall. However, it does help clarify the economic price of acting early. Future research should evaluate whether these costs were offset by reduced mortality, hospital strain, or faster recovery. More broadly, Hawaii illustrates the challenge of crisis policymaking under uncertainty.

1.7 Conclusion

This paper estimates the short-run economic consequences of Hawaii’s March 2020 mandatory quarantine using the Synthetic Historical Control (SHC) method. By leveraging the state’s own pre-treatment time series to construct credible counterfactuals, I document large negative impacts on tourism demand and related labor markets. Visitor days and hotel occupancy experienced dramatic collapses—estimated at roughly 85 and 76 percentage points below their synthetic historical benchmarks, respectively—while accommodations employment and the broader coincident economic activity index also declined sharply. These effects were statistically distinguishable from normal variation, as observed outcomes consistently fell outside 90% conformal prediction intervals. Placebo tests using a pre-pandemic intervention date confirm the model’s ability to avoid spurious detection of effects, lending credibility to the causal interpretation.

The findings highlight both the severity and the transience of the economic shock. Although the initial downturn was severe, most indicators began recovering by late 2020 and into early 2021. This pattern of front-loaded costs followed by rebound is consistent with the logic of aggressive early intervention: by severely restricting inbound travel when the virus threat was still uncertain, Hawaii absorbed a concentrated economic hit that appears to have been relatively short-lived compared to more prolonged disruptions observed elsewhere. In this sense, the results provide empirical support for the “pay now, benefit later” nature of decisive containment strategies during emerging crises.

Methodologically, this study demonstrates the value of the Synthetic Historical Control approach in settings where traditional synthetic control or difference-in-differences designs are infeasible due to the absence of suitable untreated units. Hawaii’s extreme geographic isolation and tourism-dependent economy make it a particularly challenging case for cross-sectional comparison; SHC overcomes this limitation by borrowing strength from the treated unit’s own historical dynamics. The strong pre-treatment fit, robust results across different evaluation windows, and transparent inference via conformal prediction intervals highlight SHC’s promise for evaluating one-off, systemic policy shocks—whether pandemic-related, climate-induced, or stemming from other large-scale disruptions.

From a public policy perspective, Hawaii’s experience offers a concrete data point in the enduring debate over balancing public health imperatives against immediate economic costs. While this analysis focuses exclusively on economic outcomes and does not quantify lives saved, hospital capacity preserved, or long-term epidemiological advantages, it clarifies the magnitude of the short-run trade-off. Policymakers facing future crises—be they novel pathogens, natural disasters, or supply-chain shocks—should weigh these front-loaded economic costs against the potential for avoiding more persistent damage from delayed or uncoordinated responses. The results also underscore the particular vulnerability of tourism-dependent economies to border and mobility restrictions, suggesting that targeted support for hospitality sectors may be warranted when similar containment measures are deployed.

Several limitations remain. First, the analysis captures only short-run effects through the end of 2020; longer-term scarring or differential recovery trajectories fall outside the scope of this

study. Second, while SHC provides a credible counterfactual under its stated assumptions, it cannot fully account for nationwide or global confounders that may have differentially affected Hawaii. Finally, this paper does not conduct a full cost-benefit analysis incorporating public health outcomes. Future work should integrate these economic estimates with epidemiological modeling to better assess the net welfare effects of early, stringent interventions.

Overall, Hawaii's "Great Isolation with Aloha characteristics" illustrates both the power and the pain of decisive action under uncertainty. By providing rigorous, methodologically innovative evidence on its economic consequences, this paper contributes to a more complete understanding of crisis response and equips policymakers with better tools for navigating the difficult trade-offs that future emergencies will inevitably demand.

CHAPTER 2

Clearing the Air: How India’s 2020 Lockdown Impacted Air Quality

2.1 Introduction

The COVID-19 pandemic prompted governments worldwide to implement non-pharmaceutical interventions (NPIs), such as mass lockdowns, to curb virus spread and protect public health systems. India’s nationwide lockdown, beginning on March 25, 2020, was among the strictest globally since the Wuhan lockdown in China, halting mobility, industry, and construction for 1.3 billion people. Early research focused on virus outcomes (Yang, 2021; Bayat et al., 2020) and economic impacts (Gong et al., 2024; Wu et al., 2023), but lockdowns also likely influenced air quality, a critical public health factor. Popular media reported improved air quality during lockdowns, later confirmed by studies documenting reductions in pollutants such as PM_{2.5} (Watts and Kommenda, 2020; Venter et al., 2020; Kumari and Toshniwal, 2021). Given that India has some of the most polluted cities in the world, understanding how large-scale interventions affect pollution levels is of great interest. However, many studies are descriptive and do not estimate counterfactual scenarios—what air quality would have been without lockdowns—limiting their policy relevance, or rely on strict assumptions for identification (Milman and Uteuova, 2025).

This study quantifies the causal impact of India’s 2020 national lockdown on PM_{2.5} levels nationally and in major cities using the Synthetic Historical Control (SHC) method (Chen et al., 2024). Unlike studies that rely on untreated comparison groups—which are unviable

in the COVID-19 context—or on covariate conditioning, I develop an extension of SHC that accommodates imperfect pre-treatment fit. By constructing counterfactuals from historical data with flexible, non-parametric methods, SHC isolates the lockdown’s effects without requiring unexposed control units. In addition to the national analysis, this study examines Delhi, Mumbai, Bangalore, and Kolkata—megacities consistently among the world’s most polluted, and major hubs of finance and commerce. Embedding these cases in a unified causal framework moves beyond simple pre–post comparisons and addresses a key policy question: what happens to air pollution when a country of India’s scale implements a nationwide shutdown? These findings can inform sustainable air quality management in India and provide lessons for urban planning and climate change mitigation globally.

The rest of the paper is organized as follows. Section 2.2.1 provides background on air pollution in India, highlighting its health and economic costs. Section 2.2.2 summarizes existing literature and methodological gaps. Section 2.2.2.2 outlines the empirical strategy, including SHC and the augmented version proposed here. Section 2.3 details data collection and processing.

2.2 Background and Previous Work

2.2.1 Policy Context

Air pollution is one of India’s most pressing environmental and public health challenges. This section provides context on pollutant sources, exposure levels, health and economic impacts, and how the COVID-19 lockdown created a natural experiment for studying air quality.

2.2.1.1 Sources and Severity of Air Pollution

Fine particulate matter (PM_{2.5}), which penetrates deep into the lungs and bloodstream, is a primary pollutant of concern. Major contributors include vehicular emissions, industrial activities, construction dust, biomass burning for cooking and heating, and agricultural residue burning, particularly in northern states during winter. Natural factors such as dust storms and seasonal weather patterns exacerbate pollution levels. Rapid urbanization and population growth have intensified exposure, affecting over 1.4 billion people.

Recent data highlight the severity: in 2024, India's average annual PM_{2.5} concentration was $50.6 \mu\text{g}/\text{m}^3$, far exceeding the WHO guideline of $5 \mu\text{g}/\text{m}^3$ ([Statista, 2025](#)). Regional variation is significant—cities in the Indo-Gangetic Plain, like Delhi, often exceed $100 \mu\text{g}/\text{m}^3$ during peak winter, while southern cities such as Chennai frequently surpass both WHO and national standards ([Bhuyan et al., 2025](#); [Singh et al., 2021](#)).

2.2.1.2 Health Impacts

The health consequences of chronic PM_{2.5} exposure are substantial. Long-term exposure increases the risk of respiratory and cardiovascular diseases, including COPD, asthma, lung cancer, stroke, and heart disease ([Joshi et al., 2025](#)). In India, air pollution causes roughly 1.5 million premature deaths annually, with each $10 \mu\text{g}/\text{m}^3$ rise in PM_{2.5} linked to an 8.6% increase in mortality ([Jaganathan et al., 2024](#)). Vulnerable populations, namely children, the elderly, and those with weaker immune systems, are most affected. Fine particulate pollution reduces average life expectancy by 5.3 years compared to WHO guidelines, with residents in

Delhi facing up to a 10-year reduction ([BBC, 2022](#)). Short-term PM_{2.5} spikes also increase daily death rates by 1.42–3.57% per 10 $\mu\text{g}/\text{m}^3$ ([de Bont et al., 2024](#)).

2.2.1.3 Economic Impacts

Air pollution imposes significant economic costs through healthcare expenditures, lost labor productivity, and impacts on sectors such as tourism and agriculture. Estimates suggest annual losses of USD 95 billion, roughly 1.36% of GDP ([India State-Level Disease Burden Initiative Air Pollution Collaborators, 2020](#); [Gupta and Damani, 2021](#)). In the Delhi NCR, productivity declines and reduced tourist arrivals contribute substantially to these costs ([Dalberg Advisors, 2021](#); [Our Common Air Commission, 2024](#)). Efforts like the National Clean Air Programme (NCAP), launched in 2019, aim to reduce PM_{2.5} and PM₁₀ concentrations by 20–30% in 131 non-attainment cities by 2024–25 ([Kawano et al., 2025](#)).

2.2.1.4 The COVID-19 Lockdown as a Natural Experiment

The 2020 nationwide COVID-19 lockdown provides a unique opportunity to study rapid air quality improvements. Imposed on March 25, the lockdown halted transportation, industrial operations, and construction, leading to reductions in PM_{2.5} by 31–43%, with some cities experiencing drops up to 50% ([Saleem et al., 2024](#); [Nigam et al., 2021](#); [Mishra et al., 2021](#)). In megacities like Delhi and Mumbai, the Air Quality Index (AQI) improved by as much as 58%, primarily due to lower vehicular and industrial emissions ([Zhang et al., 2020](#)).

Although temporary, these reductions highlight the potential of targeted interventions. PM_{2.5} reductions were most pronounced in early lockdown phases, averaging 30–50% across

monitored sites, with occasional spikes due to meteorology (Hu et al., 2022; Goel et al., 2021; Wang et al., 2020). Other pollutants, including NO₂ and PM₁₀, also decreased, consistent with global lockdown trends (Venter et al., 2020). Understanding these dynamics is crucial for causal evaluation and future policy planning.

2.2.1.5 Lockdown Policy Phases

India’s lockdown, enacted under the Disaster Management Act of 2005, unfolded in multiple phases. Phase 1 (March 25–April 14) involved a complete nationwide shutdown, prohibiting non-essential movement, closing factories, and suspending public transport. Phase 2 (April 15 onwards) allowed a gradual reopening of essential services and agriculture, with color-coded zoning based on infection rates (Deshpande, 2022; Ranjan et al., 2020). While primarily a public health measure, the lockdown served as a large-scale quasi-experiment in emission control, revealing the disproportionate contribution of human activities to PM_{2.5} levels.

2.2.2 Literature Review

2.2.2.1 Methodological Aspects of Prior Literature

To contextualize our approach, we review existing studies on lockdown-related air quality changes and highlight key methodological gaps. Early studies primarily relied on descriptive pre–post comparisons or satellite imagery without causal adjustment (Mahato et al., 2020; Pathakoti et al., 2021; Srivastava et al., 2020; He et al., 2020; Wang et al., 2020). These approaches document reductions in pollutants but cannot answer counterfactual questions—what would air quality have been absent the lockdown? Some studies have applied causal

models. For example, [Heffernan et al. \(2025\)](#) use causal machine learning and interrupted time series (ITS) methods to find reductions in NO₂ levels. ITS was applied in Italy ([Cameletti, 2020](#)), finding no meaningful changes in air pollutants. Augmented Synthetic Control methods were used for Wuhan ([Cole et al., 2020](#)), showing a 63% reduction in NO₂. Despite these examples, a recent systematic review finds that most lockdown–air pollution studies remain descriptive or non-causal ([Silva et al., 2022](#)), limiting their policy relevance.

2.2.2.2 Gaps and Contribution of This Study

A central gap is the lack of causal designs for national-scale interventions. Many studies use simple pre–post t-tests ([Goel et al., 2021](#)), graphical analyses ([Kumari and Toshniwal, 2020](#); [Benchrif et al., 2021](#)), year-over-year comparisons ([Le Quéré et al., 2020](#)), or OLS regressions ([Pal et al., 2022](#)). These methods cannot fully account for confounding trends or temporal patterns, reducing confidence in policy-relevant estimates. SCM improves on this by constructing a counterfactual from untreated donor units as a weighted average ([Ben-Michael et al., 2021a](#)). However, SCM is limited when studying national policies, where no valid donor pool exists, given that SCM requires untreated control units in order to be viable. Similarly, causal machine learning approaches ([Heffernan et al., 2025](#)) rely on strong ignorability assumptions that may not hold for air pollution data with unmeasured year-specific factors. In contrast, the SHC method avoids these limitations by constructing counterfactuals directly from historical patterns, without assuming complete covariate measurement. This study extends SHC to an Augmented SHC (ASHC) framework, allowing limited extrapolation to improve pre-treatment fit while maintaining stability. By applying ASHC to India’s national

and city-level PM2.5 data, this work provides the first rigorous causal estimates of the 2020 lockdown’s air quality impacts in a country of India’s scale.

Methodology

Below I describe the Synthetic Historical Control (SHC) methodology from [Chen et al. \(2024\)](#). I then extend this framework to settings where pre-treatment fit is poor, yielding the Augmented Synthetic Historical Control (ASHC). The ASHCM is applied separately to the national average of PM2.5 for India, as well as Delhi, Mumbai, Bangalore, and Kolkata.

2.2.3 The Synthetic Historical Control Method

SHC estimates the causal effect of an intervention by constructing counterfactual outcomes from historical segments of the treated unit itself, rather than from contemporaneous control units. This makes it well-suited for national-scale interventions such as India’s 2020 lockdown, where no valid donor pool of untreated countries exists. Before presenting the estimator, I state the assumptions that guarantee SHC’s validity.

2.2.3.1 Assumptions

Assumption 4 (Error Properties).

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] < \infty \quad \forall t \in \mathcal{T}. \quad (2.1)$$

This ensures that noise is well-behaved. The mean-zero condition rules out systematic bias, while the finite variance condition prevents extreme fluctuations.

Assumption 5 (Latent Smoothness and Matching).

$$\tilde{y}(t) \in C^{H+1}, \quad H \geq 3, \quad \sup_t \left| \frac{d^k}{dt^k} \tilde{y}(t) \right| < \infty \quad \forall k \leq H + 1, \quad (2.2)$$

and there exists

$$\mathbf{w}^* \in \Delta^{N-1} \quad \text{such that} \quad \mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre} \mathbf{w}^*. \quad (2.3)$$

Smoothness ensures that the latent process can be approximated by a finite expansion. The matching condition guarantees that some convex combination of historical donor segments can reproduce the evaluation period. This parallels the “existence of weights” assumption in the SCM literature ([Abadie et al., 2010](#); [Amjad et al., 2018](#)). Importantly, persistent factors such as meteorology are implicitly matched so long as they appear in the historical trajectory.

Assumption 6 (Feasibility and Stability).

$$\exists \mathbf{w} \in \Delta^{N-1} \text{ sparse, such that } \mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre} \mathbf{w}, \quad \widetilde{\mathbf{Y}}_{pre}^\top \widetilde{\mathbf{Y}}_{pre} \succ 0. \quad (2.4)$$

Sparsity means that only a small number of donor segments are relevant. Positive definiteness prevents collinearity and ensures stable weights. In plain terms, SHC gives us a valid estimate of the lockdown’s impact as long as three conditions hold. First, the random “noise” in pollution measurements behaves well — it averages out and does not swamp the real signal. Second, the history of pollution before the lockdown is smooth and informative enough that past patterns can be combined to mimic what would have happened in the absence of the

lockdown. This means that unobserved influences like weather are implicitly accounted for, so long as they leave a stable footprint in the historical data. Third, the relevant pieces of history we draw on to form the counterfactual are not redundant, so the weights we place on them are stable rather than erratic. Under these conditions, SHC can reliably construct the missing counterfactual trajectory and deliver a valid estimate of the average treatment effect of the lockdown.

2.2.3.2 Estimator

Let $\mathcal{T} = \{1, \dots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \dots, T_0\}$. Let $\mathbf{y} = (y_1, \dots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \quad d_t = \mathbf{1}\{t > T_0\}. \quad (2.5)$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\hat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as

$$\hat{\alpha}_t = y_t - \hat{y}_t^0, \quad t \in \mathcal{T}_2, \quad (2.6)$$

with the average treatment effect on the treated (ATT) defined as

$$\widehat{\text{ATT}} = \frac{1}{n} \sum_{t \in \mathcal{T}_2} \hat{\alpha}_t. \quad (2.7)$$

The implementation of SHC begins by smoothing the pre-treatment outcome series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, yielding a smooth trajectory $\tilde{\mathbf{y}}$ (Fan and Gijbels, 1992). From this smoothed series, we construct the donor matrix by extracting N overlapping segments, each of length m .

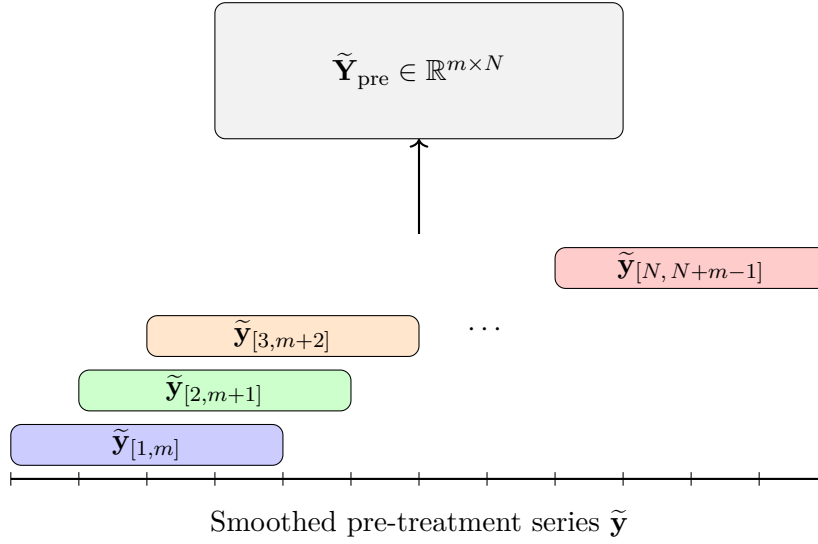


Figure 2.1: Constructing the donor matrix $\tilde{\mathbf{Y}}_{\text{pre}}$ by sliding overlapping windows of length m across the smoothed pre-treatment series $\tilde{\mathbf{y}}$. Each colored segment corresponds to a column in the donor matrix.

The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \quad \text{s.t. } \mathbf{w} \geq 0, \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with small eigenvalues. The second term penalizes directions of low variance, improving numerical stability when donor segments are collinear.

In practice, only a few donor segments contribute meaningfully to the reconstruction of $\mathbf{y}_{\text{eval}}$. To identify these, I use the forward selection algorithm described by [Chen et al. \(2024\)](#). Applying the resulting weight vector $\mathbf{w}^*$ to the aligned post-treatment donor matrix produces $\hat{\mathbf{y}}_{\text{post}}^0$, the SHC estimate of the untreated counterfactual trajectory.

A further advantage of SHC in this setting is that it naturally accommodates nonlinear dynamics in the pre-treatment pollution series. Because SHC begins with a local linear smoother, short-term fluctuations and seasonal nonlinearities (many of which are driven by meteorology such as temperature inversions, wind, or rainfall) are already captured in the donor segments. In contrast, standard interrupted time series designs often require explicitly specifying and fitting complex parametric models such as SARIMA to handle the same effects. Thus, identification in SHC setting rests only on the modest assumption of local smoothness and weight matching, with the primary tuning choice being the bandwidth of the kernel smoother. This makes the method flexible enough to capture weather-driven variation while remaining relatively simple to implement in practice.

2.2.4 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the

solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization. Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weight vector.

The ASHC estimator solves

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \underbrace{\left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2}_{\text{Fit Term}} + \underbrace{\varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2}_{\text{Low-variance term}} + \underbrace{\lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2}_{\text{Proximity term}}, \quad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1. \quad (2.8)$$

As we can see, the first two terms are essentially unchanged from the original SHC method.

The third term in the ASHC objective,

$$\lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. However, the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\text{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against

the potential gains in fit from extrapolation. The ASHC counterfactual is obtained as

$$\hat{\mathbf{y}}_{\text{post}}^{0,\text{ASHC}} = \widetilde{\mathbf{Y}}_{\text{post}} \mathbf{w}^{\text{ASHC}}, \text{ with treatment effects estimated analogously to SHC.}$$

2.2.5 Bandwidth and Regularization Parameter Selection

The performance of SHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ . Both are chosen via cross-validation. First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\text{pre}} = (y_1, \dots, y_{T_0})^\top$ denote the pre-treatment series. For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\hat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j - i) = \exp\left(-\frac{1}{2} \left(\frac{j-i}{h}\right)^2\right),$$

normalized so that $\sum_{j \neq i} K_h(j - i) = 1$. Formally, for each $i \in \{1, \dots, T_0\}$,

$$\hat{\beta}(h)^{(-i)} = \arg \min_{\beta_0, \beta_1} \sum_{j \neq i} K_h(j - i) \left(y_j - \beta_0 - \beta_1(j - i) \right)^2,$$

and the leave-one-out prediction is $\hat{y}_i^{(-i)}(h) = \hat{\beta}_0(h)^{(-i)}$. The LOOCV error is $\text{CV}(h) = \frac{1}{T_0} \sum_{i=1}^{T_0} \left(y_i - \hat{y}_i^{(-i)}(h) \right)^2$. The optimal bandwidth is then chosen as $h^* = \arg \min_{h \in \mathcal{H}} \text{CV}(h)$, where $\mathcal{H}$ is a prespecified candidate grid. This is chosen before the SHC estimates weights for any of the time periods, and this carries over to the ASHC method.

Next we discuss λ , exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w}; \lambda) = \left\| \mathbf{y}_{\text{train}} - \widetilde{\mathbf{Y}}_{\text{train}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2, \quad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1. \quad (2.9)$$

Let the evaluation window $\mathcal{T}_{\text{eval}}$ be split into a training set $\mathcal{T}_{\text{train}}$ and a validation set $\mathcal{T}_{\text{val}}$, with corresponding subvectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\text{train}}, \widetilde{\mathbf{Y}}_{\text{val}}$.

We construct a logarithmically spaced grid of candidate values

$$\Lambda = \left\{ \lambda_j : \lambda_j = \lambda_{\max} \rho^j, \quad j = 0, 1, \dots, n_\lambda - 1 \right\}, \quad (2.10)$$

where

$$\lambda_{\max} = \sigma_{\max}^2, \quad \lambda_{\min} = \lambda_{\max} \cdot \kappa, \quad \rho = \left(\frac{\lambda_{\min}}{\lambda_{\max}} \right)^{\frac{1}{n_\lambda - 1}}. \quad (2.11)$$

Here $\sigma_{\max}$ denotes the largest singular value of $\widetilde{\mathbf{Y}}_{\text{train}}$, and $\kappa > 0$ is a user-chosen constant controlling the lower bound of the grid.¹ Thus Λ spans values from $\lambda_{\max}$ (very strong shrinkage toward $\mathbf{w}^{\text{SHC}}$) down to $\lambda_{\min}$ (minimal shrinkage).

For each $\lambda \in \Lambda$, the estimator is

$$\widehat{\mathbf{w}}(\lambda) = \arg \min_{\mathbf{w} \in \mathbb{R}^N} Q(\mathbf{w}; \lambda), \quad (2.12)$$

¹In practice, we set $\kappa = 10^{-8}$, ensuring that the grid spans from strong shrinkage ($\lambda_{\max}$) to nearly unpenalized fits ($\lambda_{\min}$).

and the validation error is

$$\text{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\text{val}}|} \left\| \mathbf{y}_{\text{val}} - \widetilde{\mathbf{Y}}_{\text{val}} \widehat{\mathbf{w}}(\lambda) \right\|_2^2. \quad (2.13)$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg \min_{\lambda \in \Lambda} \text{Val}(\lambda). \quad (2.14)$$

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

2.3 Data

My data on PM2.5 are derived from [Batra \(2025\)](#), who keep data collected from the Socioeconomic High-resolution Rural-Urban Geographic Data Platform for India (SHRUG). We have the full district-level PM2.5 matrix:

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,T} \end{bmatrix}, \quad N = 635, \ T = 312$$

where rows are districts ($i = 1, \dots, N$) and columns are monthly population weighted averages of PM2.5 observations ($t = 1, \dots, T$). For the national level of analysis, we compute the

population-weighted average across all rows

$$\mathbf{y}_{\text{India}} = \frac{1}{N} \sum_{i=1}^N \mathbf{Y}_{i,\cdot} \in \mathbb{R}^T.$$

This returns a single time series of the empirical mean of PM2.5 across India for the full time series. For our district-level treated units ($c \in \{\text{Delhi, Mumbai, Bangalore, Kolkata}\}$), we extract subvector(s) corresponding to the district

$$\mathbf{y}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{Y}_{i,\cdot} \in \mathbb{R}^T.$$

where S_c is the set of subdistricts belonging to district c . My final outcome is the year-over-year (YoY) growth rate of the PM25 levels: for any series $\bar{\mathbf{y}}$, the monthly YoY growth from month t relative to the same month in the previous year is

$$g_t = \frac{\bar{y}_t - \bar{y}_{t-12}}{\bar{y}_{t-12}}, \quad t = 13, \dots, T$$

so that the final treated units for analysis are

$$\mathbf{g}_{\text{India}} = [g_{13}, \dots, g_T], \quad \mathbf{g}_c = [g_{13}, \dots, g_T] \forall c.$$

This restricts the analysis to 1999–2020, corresponding to months 13 through 312 in the original dataset.

2.4 Results

Figure 2.2 presents the Synthetic Historical Control estimate for the all-India population-weighted PM2.5 series, expressed as a year-over-year growth rate. The observed series falls sharply below its synthetic-historical counterfactual following the March 2020 lockdown, indicating a pronounced short-run reduction in particulate pollution. Figure 2.3 repeats the exercise for the four megacities (Delhi, Mumbai, Bangalore, and Kolkata).

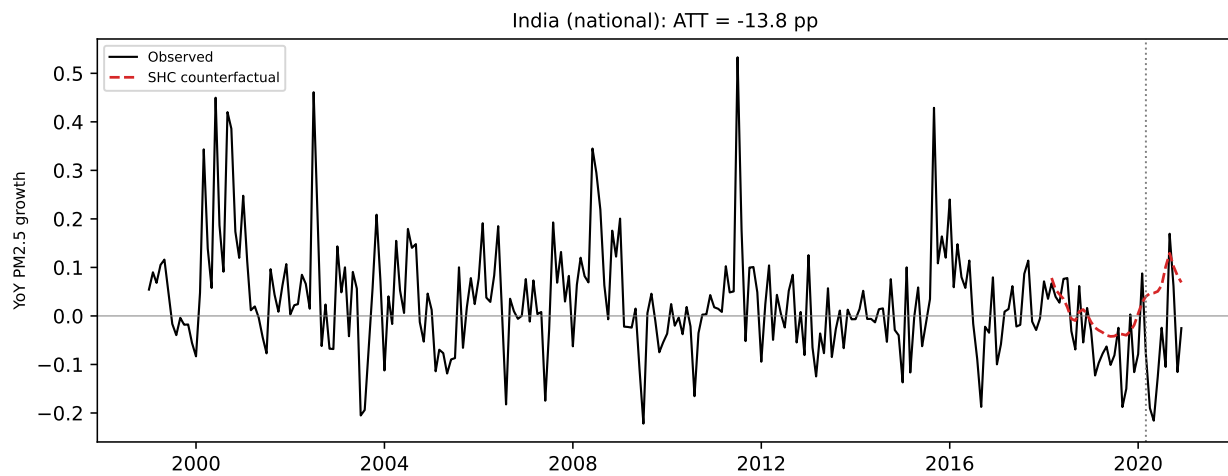


Figure 2.2: Synthetic Historical Control: all-India year-over-year PM2.5 growth, observed versus counterfactual. The dashed line marks the March 2020 national lockdown.

2.5 Robustness: Quarterly Aggregation

Monthly PM2.5 growth rates are inherently noisy: meteorology, episodic events (festivals, crop-residue burning), and measurement error all inject high-frequency variation that can exaggerate or mask any single month's estimated effect—the September 2020 rebound visible in Figure 2.2 is one such idiosyncratic spike. To confirm that the headline results are not an artifact of this monthly noise, I re-estimate every model on quarterly data: the population-

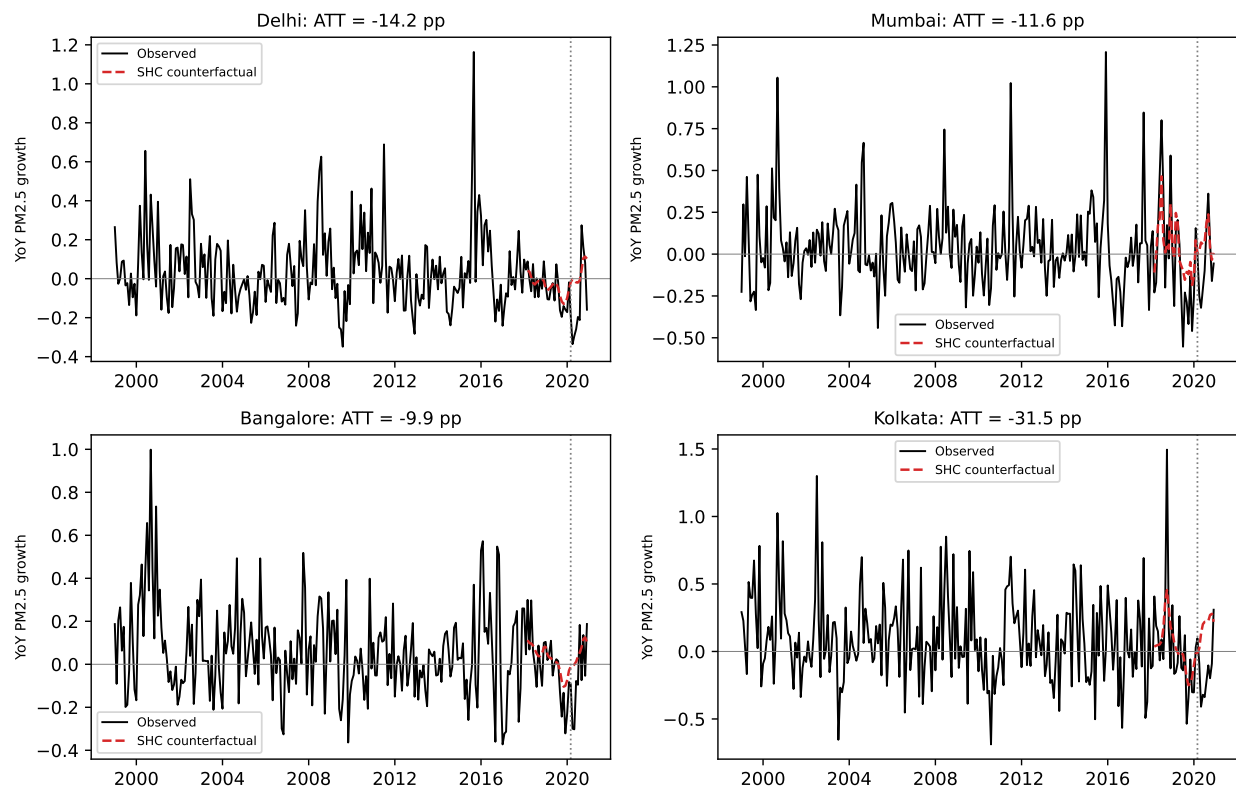


Figure 2.3: Synthetic Historical Control for the four megacities: observed versus counterfactual year-over-year PM2.5 growth.

weighted PM2.5 series is averaged into calendar quarters before the year-over-year growth rate is formed and the SHC estimator is re-run. Because the strict lockdown spanned the entire second quarter of 2020, quarterly aggregation smooths out month-to-month shocks while preserving the policy signal.

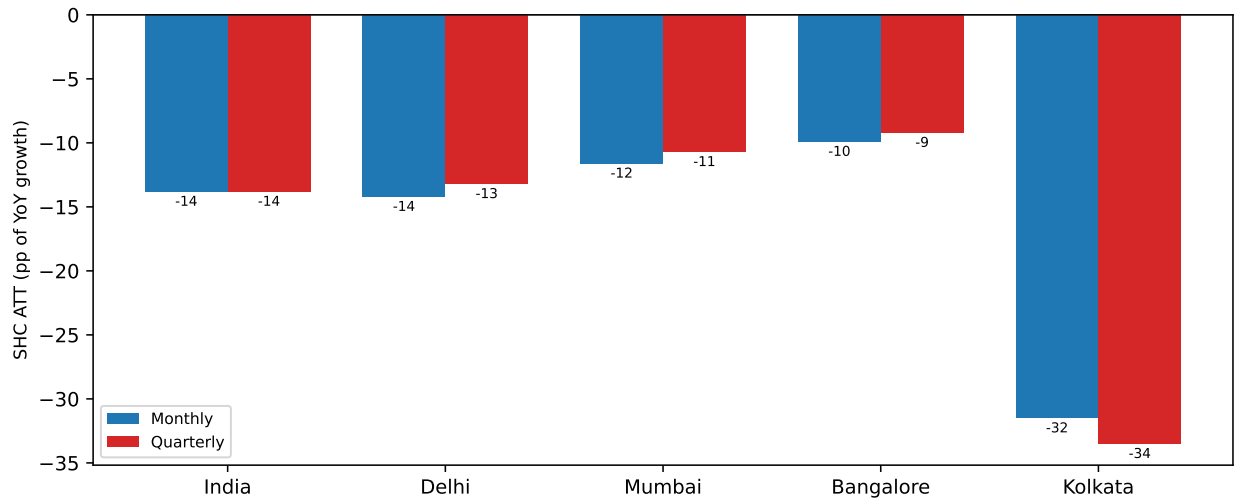


Figure 2.4: Robustness of the SHC estimates to temporal aggregation: the average treatment effect (percentage points of year-over-year PM2.5 growth) estimated at the monthly frequency versus the same estimator on quarterly data, for India and the four megacities.

Figure 2.4 compares the monthly and quarterly SHC estimates for all five units, and the two are strikingly close. Nationally, the quarterly ATT is 13.8 pp against 13.8 pp at the monthly frequency; the city estimates likewise track their monthly counterparts to within about two percentage points (Delhi 13.2, Mumbai 10.7, Bangalore 9.2, and Kolkata 33.5 pp). The cross-city ordering is preserved—Kolkata largest, Bangalore smallest—and every effect remains a substantial reduction. Because the point estimates are essentially invariant to whether the data are analyzed monthly or quarterly, the measured lockdown effect reflects a genuine, sustained shift in particulate pollution rather than a handful of anomalous months.

2.6 Discussion

Across every series, the Synthetic Historical Control (SHC) estimates in Figure 2.2 and Figure 2.3 point in the same direction: India’s 2020 lockdown sharply suppressed fine-particulate pollution. Because the outcome is the year-over-year (YoY) growth rate of PM2.5 (Section 2.3), each estimate is a change in that growth rate measured in percentage points (pp). Nationally, the average treatment effect on the treated is a 13.8 pp reduction: the synthetic-historical counterfactual implies all-India PM2.5 would have *grown* by roughly 7.3% year over year on average between March and December 2020, yet the observed series fell instead. The wedge between the two is the causal signature of the shutdown, and it dwarfs the ordinary year-to-year movement of the pre-treatment series.

The effect is concentrated exactly where the policy was strictest. Reductions are deepest during the Phase 1–2 window of April–June 2020, when mobility, industry, and construction were halted nationwide (Section 2.2.1); the largest single-month national gap occurs in May 2020, about 26.3 pp below counterfactual. From mid-summer the effect attenuates as the economy reopened, and a brief positive deviation around September 2020 appears in several series. This profile—a deep trough under the strictest restrictions followed by mean reversion—is the signature of a temporary non-pharmaceutical intervention rather than a durable shift in the emissions regime. It also underscores a strength of SHC in this setting: recurring meteorological drivers (Section 2.2.3) are absorbed into the historical donor segments, so the estimated effect is not an artifact of a single anomalous season.

City-level estimates reveal meaningful heterogeneity (Figure 2.3). The reduction is 14.2 pp in Delhi, 11.6 pp in Mumbai, 9.9 pp in Bangalore, and 31.5 pp in Kolkata. Kolkata’s effect is both the largest and the most persistent: its counterfactual was on a steep upward path (averaging about 17.8% YoY growth), so the shutdown opened an especially wide gap that endured through the autumn. Delhi shows the textbook pattern of a steep spring collapse followed by a rebound late in the year, when post-monsoon crop-residue burning and winter temperature inversions—drivers outside the lockdown’s reach—reassert themselves. Bangalore, with a lighter industrial base and a cleaner baseline, shows the smallest reduction, consistent with its pollution being comparatively less sensitive to the halt in heavy industry and construction.

These causal estimates are broadly consistent with, but conceptually distinct from, the descriptive 31–43% concentration declines reported in the lockdown literature (Nigam et al., 2021; Saleem et al., 2024): rather than comparing raw before-and-after levels, SHC benchmarks the observed series against what its own history implies should have happened. Several caveats temper interpretation. The estimates are for a transitory shock and speak to short-run responsiveness, not to a sustainable abatement path. The outcome is a growth rate, so a negative ATT denotes slower growth—here, outright decline—relative to counterfactual rather than a level reduction per se. The figures report the convex SHC estimator; the augmented variant of Section 2.2.4 is available where pre-treatment fit is poor, and the September rebound is a reminder that the design recovers net effects, including any offsetting seasonal forces—though aggregating to quarters (Section 2.5) averages out this high-frequency

variation and leaves the point estimates essentially unchanged. Finally, formal uncertainty quantification is not reported alongside these point estimates and is a natural next step.

2.7 Conclusion

This chapter provides what is, to my knowledge, the first set of causal Synthetic Historical Control estimates of the air-quality consequences of India’s 2020 national lockdown, at both the national level and for four of its largest megacities. The lockdown lowered the year-over-year growth of PM2.5 by about 13.8 pp nationally, with city effects ranging from 9.9 pp in Bangalore to 31.5 pp in Kolkata. In every case the observed pollution path fell below a counterfactual that, on its own historical momentum, would have continued to rise.

Two implications follow. First, the speed and size of the response confirm that a large share of India’s particulate burden is anthropogenic and acutely sensitive to economic activity—transport, industry, and construction—rather than fixed by geography or climate alone. Second, the rapid rebound once restrictions eased shows that one-off shocks do not deliver lasting gains: realizing the air-quality improvements glimpsed in 2020 would require sustained, structural emission controls of the sort envisioned by the National Clean Air Programme (Section 2.2.1), not episodic shutdowns.

The analysis also points to clear avenues for future work: attaching formal inference (for example, conformal prediction intervals) to the SHC point estimates, deploying the Augmented SHC of Section 2.2.4 where pre-treatment fit is weakest, extending the outcome set beyond PM2.5 to co-pollutants such as NO₂, and tracing how quickly pollution returns to its pre-

pandemic trajectory. Taken together, the results demonstrate both the public-health stakes of India's air pollution and the practical value of historical-control methods for evaluating large-scale interventions where no untreated comparison unit exists.

CHAPTER 3

Locking Away Prosperity? Evaluating the Labor Impacts of Vaccine Mandates

3.1 Introduction

The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, [Friedson et al. \(2021\)](#) and [Yang \(2021\)](#) use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, [Bayat et al. \(2020\)](#) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like [Gibson and Sun \(2023\)](#) and [Lee and Yang \(2022\)](#) document labor market costs of lockdowns, while [Ke and Hsiao \(2022\)](#) show that although strict NPIs incur short-term economic losses, those effects may be temporary.

By 2021, as vaccines became widely available, cities such as New York City (NYC), New Orleans (NOLA), Los Angeles (LA), and San Francisco (SF) turned to vaccine mandates as a less restrictive (but still controversial) tool for curbing transmission.¹ Unlike lockdowns, mandates aimed to sustain economic activity in industry while reducing health risks in

¹Henceforth, I refer to these simply as the treated units

high-contact settings. Nevertheless, critics argued that requiring proof of vaccination for indoor dining and entertainment could slow employment recovery in restaurants by alienating customers or exacerbating staffing shortages ([Dive, 2021](#)). Supporters countered that mandates would improve public safety and restore consumer confidence, potentially boosting demand and employment ([Jones, 2021](#)).

Each city's mandate differed in timing and design. NYC's *Key to NYC* program, announced in August 2021 and enforced beginning September, was the first large-scale vaccine mandate for indoor commerce in the United States ([Fortney, 2021](#)). San Francisco followed with an even stricter rule requiring full vaccination for customers and employees, effective later that same month ([Duffett, 2021](#)). New Orleans adopted a hybrid approach, permitting either proof of vaccination or a recent negative test ([Lorell, 2021](#)), while Los Angeles—by far the largest jurisdiction—implemented its policy in November ([Reyes, 2021](#)). These mandates created a small but consequential set of urban outliers in the national policy landscape, with important implications for labor markets in the leisure and hospitality sector. Precisely, the estimated effects of city-level vaccine mandates on restaurant employment suggest several potential lessons for policymakers and business stakeholders. If the mandates support employment growth, policymakers could view them as a viable tool for balancing public health and economic activity. Cities considering similar interventions might adopt comparable policies with confidence that labor markets will not be severely disrupted. Businesses could plan for future public health policies with reduced concern for employment losses in high-contact sectors. If the mandates depress employment, policymakers should weigh the trade-offs more

carefully before implementing mandates, especially in sectors sensitive to consumer demand and staffing shortages. Alternative interventions, such as targeted incentives for vaccination, improved workplace safety measures, or temporary subsidies, might be preferable to blunt employment impacts of mandates. Businesses may need mitigation strategies, including flexible staffing or remote services, to reduce vulnerability to future mandates. If the effects are heterogeneous across cities, differences in local conditions such as compliance rates, labor market tightness, or enforcement intensity should guide tailored policy decisions rather than blanket mandates. Comparative case studies could help identify which contextual factors drive positive versus negative outcomes.

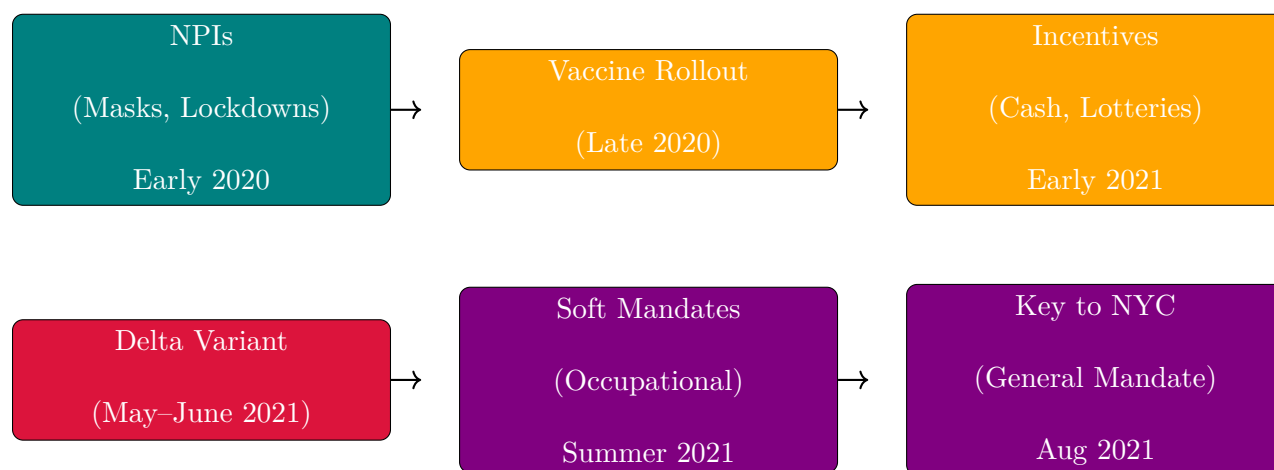
This paper evaluates the causal effects of these city-level vaccine mandates on restaurant employment. While prior research has analyzed mandates' impacts on vaccination rates and health outcomes ([Cohn et al., 2022](#); [Acharya and Dhakal, 2021](#); [Lang et al., 2023](#)), relatively little is known about their labor market consequences, especially in industries dependent on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, the analysis adopts a synthetic control approach to estimate the counterfactual trajectory of restaurant employment absent the mandates. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct synthetic controls for each mandate city's year-over-year growth in full-service restaurant employment. The donor pool consists of metropolitan areas that did not implement similar mandates during the same period.²

The rest of the paper is organized as follows. Section [3.2](#) provides background on the evolution

²For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

of pandemic interventions and the city-level vaccine mandates. Section 3.3 describes the employment data and outcome construction. Section 3.4 outlines the empirical strategy, including the synthetic control methodology. Section 3.5 presents the main findings and robustness checks. Section 3.6 interprets the results and discusses limitations. Section 3.7 concludes.

3.2 Policy Background



Timeline of major pandemic policy phases

3.2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so,

with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action ([Hansen and Mano, 2023](#); [Chernozhukov et al., 2021](#); [Mitze et al., 2020](#)). While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they still provoked political backlash due to their perceived disruptive nature (especially in the United States).

3.2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. Unlike non-pharmaceutical interventions, which sought to reduce transmission primarily by altering citizens behavior and limiting contact (and therefore the spread of the virus), vaccine-based interventions operate directly on biological susceptibility. NPIs such as lockdowns, mask mandates, and border closures work by creating external barriers to viral spread, often imposing economic or social costs in the process. Vaccine mandates, by contrast, rely on an individual's immune response to prevent infection or reduce severity, aiming to achieve population-level protection without requiring broad restrictions on mobility or commerce. In principle, this allows policymakers to target public health objectives more directly, potentially mitigating the trade-offs between controlling the virus and maintaining economic activity. Nevertheless, vaccine-based mandates introduce new challenges, including compliance, equity, and political acceptability, which

differ in character from those associated with NPIs.

For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older adults and (often mandated for) healthcare workers ([Okpani et al., 2024](#)). However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination ([Barber and West, 2022](#); [Robertson et al., 2021](#)). These included direct cash payments, vaccine lotteries (e.g., Ohio’s “Vax-a-Million” from May to June ([Fuller et al., 2022](#))), and smaller symbolic giveaways. One paper dubbed this the era of “donuts, drugs, booze, and guns,” reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements ([Tinari and Riva, 2021](#)).

Despite their creativity, most incentives had modest or short-lived effects ([Saban et al., 2021](#); [Walkey et al., 2021](#); [Khazanov et al., 2023](#)). Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment ([American Medical Association et al., 2021](#)). Over the summer, local governments and organizations expanded this approach ([American Medical Association et al., 2021](#); [Al Jazeera, 2021](#)).

3.2.3 City-Level Vaccine Mandates

The summer 2021 Delta surge spurred several major U.S. cities to enact vaccine verification requirements for indoor restaurants and leisure venues. Each city’s approach differed slightly in timing and scope, but all shared the common goal of linking access to indoor leisure activities with proof of vaccination. Together, the treated units formed the clearest urban experiments in the use of vaccine mandates to manage both public health risks and consumer confidence in the hospitality sector.

New York City’s “Key to NYC” program was the first such mandate in the United States. Announced on August 3, 2021, formalized on August 16, and enforced beginning September 13, it required proof of at least one vaccine dose to enter indoor dining, fitness centers, and entertainment venues ([Benveniste, 2021](#)). Importantly, the rule applied not only to patrons but also to employees of covered establishments. This dual requirement meant that the policy had the potential to affect restaurant employment through both the supply side, if workers resisted vaccination or exited the industry, and the demand side, if consumer traffic shifted in response to the mandate. The city threatened fines for noncompliance and required businesses to post signage and maintain written protocols, though the intensity of inspections varied. Large chains tended to comply quickly ([Messenger, 2021](#)), while smaller establishments voiced concern about administrative burdens and customer pushback. Several lawsuits challenged the mandate, but city officials argued that rising vaccination rates demonstrated its effectiveness ([Wiener-Bronner, 2021](#)). San Francisco followed closely, enacting its mandate on August 20, 2021. Unlike New York, the city required full vaccination rather than just one dose, and

did not allow a testing alternative ([San Francisco, 2021](#); [Duffett, 2021](#)). This made San Francisco's policy the strictest among the four cities. The requirement extended to both customers and employees in indoor restaurants, bars, and entertainment venues. Enforcement included spot checks and the threat of fines, with compliance reported to be widespread among larger businesses in the city's dense urban core. As in New York, officials promoted the policy as a tool for stabilizing consumer confidence in public spaces, even as smaller businesses expressed anxiety about turning customers away.

New Orleans became the first major city in the South to adopt a mandate, announcing its policy on August 16, 2021. The city's approach was a hybrid model: individuals could gain entry to indoor restaurants and venues either by showing proof of at least one vaccine dose or by presenting a recent negative COVID test ([Aya and Riess, 2021](#)). This flexibility reflected local political and cultural dynamics but also introduced more administrative complexity for restaurants. Enforcement began within weeks ([Just, 2021a,b](#)), with inspections carried out by city officials. Restaurant owners reported mixed reactions, with some noting relief that the policy might make indoor dining safer, while others expressed concern about deterring unvaccinated tourists and residents ([Chavez, 2021](#)). Los Angeles entered the field later, passing its ordinance in October 2021 with enforcement beginning November 4. The policy was sweeping in scope, covering indoor restaurants, gyms, theaters, salons, and other leisure establishments. By the time it came into effect, public debate had already been shaped by the experience of New York and San Francisco, and Los Angeles became the largest jurisdiction in the nation to implement a vaccine passport ([Solis and Branson-Potts, 2021](#)).

Although the four cities differed in timing and strictness, they shared important common features. Each policy was applied citywide, targeted indoor leisure and hospitality settings, and created new compliance responsibilities for businesses. They also diverged sharply from the policy environments of states such as Texas and Florida, which prohibited vaccine requirements altogether. The result was a small cluster of urban outliers using vaccine mandates as a central public health intervention in 2021. These institutional details carry important implications for research design. Because the mandates covered both patrons and employees in most cases, they may have constrained labor supply by excluding some workers while simultaneously influencing labor demand by shaping customer traffic. Enforcement intensity and public perception varied, but each policy represented a reasonably sharp and well-documented intervention, announced and enforced on a specific timeline. This combination makes them suitable for empirical evaluation. Comparing employment outcomes in these cities against synthetic control groups of metropolitan areas without mandates offers a credible way to estimate causal effects on restaurant employment. The staggered adoption across cities and the variation in stringency further enrich the research design, providing multiple natural experiments within a single policy family.

3.2.4 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher.

However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3.3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and, by extension, the Bureau of Labor Statistics, spanning January 1991 through December 2022, yielding $T = 384$ monthly observations per unit. Let $\mathcal{J} = \{1, \dots, J\}$ denote the set of treated units, corresponding to New York City, San Francisco, New Orleans, and Los Angeles. For treated unit $j \in \mathcal{J}$, the pre-treatment period is defined as

$$\mathcal{T}_1^{(j)} = \{t \in \mathbb{N} : t \leq T_0^{(j)}\},$$

where $T_0^{(j)}$ is the final period before the mandate in city j . Similarly, the post-treatment period is

$$\mathcal{T}_2^{(j)} = \{t \in \mathbb{N} : t > T_0^{(j)}\}.$$

Our analysis focuses on two related outcome sets: (1) full-service restaurant employment for the treated cities and 30 additional MSAs, and (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022.

Let e_{jt} denote employment in unit j at month t , and define the year-over-year growth rate in percentage points as

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \quad j = 1, \dots, N, \quad t = 13, \dots, T,$$

using smoothed and seasonally adjusted values. Outcomes for each unit are collected into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top$.

For each treated city $j \in \mathcal{J}$, a separate synthetic control is estimated using a donor pool

$$\mathcal{N}_0^{(j)} = \mathcal{N} \setminus (\{j\} \cup \mathcal{J} \setminus \{j\}),$$

which excludes the treated unit itself as well as the other treated units. The treatment indicator is now

$$d_{jt} = \mathbf{1}\{j \in \mathcal{J} \wedge t > T_0^{(j)}\},$$

so observed outcomes satisfy

$$y_{jt} = (1 - d_{jt})y_{jt}(0) + d_{jt}y_{jt}(1),$$

with causal effects

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

The post-treatment average treatment effect on the treated (ATT) is

$$\text{ATT} = \frac{1}{J} \sum_{j \in \mathcal{J}} \frac{1}{|\mathcal{T}_2^{(j)}|} \sum_{t \in \mathcal{T}_2^{(j)}} \tau_{jt},$$

averaging first over the post-treatment months within each city and then across all treated cities.

3.4 Econometric Methodology

The four city-level mandates form a staggered-adoption panel with three treated units (after dropping New Orleans for data availability) and a donor pool of metropolitan areas that did not enact comparable indoor vaccine-verification policies. Treatment timing differs across

cities, and the cities themselves — large, coastal, tourism-dependent metropolitan economies — often lie near or outside the convex hull of the donor pool. Classical synthetic control (Abadie et al., 2010), which fits a separate convex combination of donors to each treated unit, fails on both counts: it tolerates only one adoption date at a time and produces poor pre-treatment fit whenever the treated unit is geometrically extreme. I therefore deploy two estimators specifically designed for staggered adoption with potentially poor unit-level fit: the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), with the event-study disaggregation of Ciccia (2024), and the partially pooled synthetic control (PPSCM) estimator of Ben-Michael et al. (2021b). The two methods share the synthetic-control philosophy of fitting a counterfactual by data-driven weighting but differ in what they weight and how they trade off competing imbalance criteria; reporting both lets the estimated effect be triangulated rather than identified by a single modeling choice.

3.4.1 *Synthetic Difference-in-Differences*

SDID generalizes the canonical SCM idea by introducing time weights alongside unit weights and embedding both in a weighted two-way fixed-effects regression. For a given cohort a of treated units (here, one cohort per treated city), unit weights $\hat{\omega}^{sdid}$ are chosen on the simplex to make the donor pool’s pre-treatment trajectory parallel to that of the treated unit, and time weights $\hat{\lambda}^{sdid}$ are chosen on the simplex to make the donor pool’s pre- and post-treatment average outcomes agree apart from a constant. Concretely, $\hat{\omega}^{sdid}$ solves

$$\hat{\omega}_0, \hat{\omega}^{sdid} = \underset{\omega_0 \in \mathbb{R}, \omega \in \Omega}{\operatorname{argmin}} \sum_{t=1}^{T_0} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{it} - \frac{1}{N_{tr}} \sum_{i \in \mathcal{I}^a} Y_{it} \right)^2 + \zeta^2 T_0 \|\omega\|_2^2,$$

with the simplex constraint $\boldsymbol{\omega} \geq \mathbf{0}$, $\mathbf{1}^\top \boldsymbol{\omega} = 1$, and ζ a ridge term calibrated to the standard deviation of first-differenced donor outcomes (Arkhangelsky et al., 2021). The time-weight problem is analogous, with no ridge penalty. The cohort-specific SDID estimator is then a weighted double-difference:

$$\hat{\tau}_a^{sdid} = \frac{1}{T_{tr}^a} \sum_{t=a}^T \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \hat{\omega}_i Y_{it} \right) - \sum_{t=1}^{a-1} \hat{\lambda}_t \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \hat{\omega}_i Y_{it} \right),$$

i.e., the post-treatment gap between treated and synthetic control minus a time-weighted baseline computed on the pre-treatment window. Cohort-specific effects are aggregated into an overall ATT via the treated-unit-weighted scheme of Clarke et al. (2023), equivalent to the event-study aggregation of Ciccia (2024) when applied with uniform horizon weights.

The estimator’s appeal in this setting is twofold. First, the time weights $\hat{\boldsymbol{\lambda}}^{sdid}$ implicitly down-weight pre-treatment months whose outcome dynamics look unlike the post-treatment window. In a panel that straddles the COVID-induced labor-market trough of 2020, this is non-trivial: months in which restaurant employment fell catastrophically below trend are mechanically down-weighted relative to months in which year-over-year growth was more typical, so the synthetic control is not anchored to a counterfactual dominated by the worst of the pandemic. Second, the two-way fixed effects absorb common time-invariant unit shifts and common contemporaneous shocks, weakening the parallel-trends assumption that DiD strategies usually require. Inference is conducted by placebo permutation, in which each donor MSA is iteratively reassigned as a pseudo-treated unit, SDID is re-fit, and

the resulting ATT distribution is used to form a standard error and a two-sided p-value $p = (\#\{|\hat{\tau}^*| \geq |\hat{\tau}|\} + 1)/(B + 1)$ with $B = 500$ placebo replications.

3.4.2 Partially Pooled Synthetic Control

Partially pooled SCM (Ben-Michael et al., 2021b) addresses the same staggered-adoption problem from a different angle. Rather than augmenting SCM with time weights and fixed effects, it generalizes SCM by jointly estimating a weight matrix $\mathbf{\Gamma} \in \mathbb{R}^{N_0 \times J}$ across all treated units and balancing two distinct pre-treatment imbalance criteria. Let $\mathbf{y}_j^{\mathcal{T}_1}$ denote treated unit j 's pre-treatment outcome vector and $\mathbf{Y}_0^{\mathcal{T}_1}$ the aligned donor matrix. Define

$$q_{\text{sep}}(\mathbf{\Gamma})^2 = \frac{1}{J} \sum_{j=1}^J \frac{1}{L} \left\| \mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \boldsymbol{\gamma}_j \right\|_2^2,$$

$$q_{\text{pool}}(\mathbf{\Gamma})^2 = \frac{1}{L} \left\| \frac{1}{J} \sum_{j=1}^J \left(\mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \boldsymbol{\gamma}_j \right) \right\|_2^2,$$

where the first quantity averages pre-treatment fit error across treated units (penalizing weak unit-level fits) and the second averages residuals first and then computes error (penalizing weak fit to the average treated unit). The two measures are not redundant: $q_{\text{pool}} \leq q_{\text{sep}}$ always, and the gap is large when treatment timing or unit characteristics vary across the treated group. PPSCM solves a normalized convex combination,

$$\hat{\mathbf{\Gamma}} = \underset{\mathbf{\Gamma} \in \Delta_{\text{scm}}^J}{\text{argmin}} \nu \cdot \tilde{q}_{\text{pool}}(\mathbf{\Gamma})^2 + (1 - \nu) \cdot \tilde{q}_{\text{sep}}(\mathbf{\Gamma})^2 + \lambda \|\mathbf{\Gamma}\|_F^2,$$

with $\tilde{q}_{\text{sep}}$ and $\tilde{q}_{\text{pool}}$ normalized by their values at the $\nu = 0$ (separate-SCM) baseline. The pooling intensity $\nu \in [0, 1]$ governs the trade-off: $\nu = 0$ recovers separate SCM on each treated city followed by averaging, $\nu = 1$ recovers fully pooled SCM in which the treated cities are averaged first and a single synthetic control is fit to the average, and intermediate ν moves smoothly along the convex frontier of the two imbalances. I select ν data-adaptively as the value that minimizes the sum $\tilde{q}_{\text{sep}}(\mathbf{\Gamma}_\nu) + \tilde{q}_{\text{pool}}(\mathbf{\Gamma}_\nu)$ over a 21-point grid on $[0, 1]$ — the balance-frontier knee that [Ben-Michael et al. \(2021b\)](#) recommend in the absence of error-bound parameters. The Frobenius-norm penalty $\lambda \|\mathbf{\Gamma}\|_F^2$ stabilizes the weights and is set to a small numerical value to break degeneracies without materially shifting the fit. I report results both with and without the paper’s “intercept shift” extension, which subtracts each unit’s pre-treatment mean before fitting; the toggle has minimal effect on the headline estimate, which I treat as a passed robustness check on the demeaning choice.

Inference for PPSCM uses leave-one-treated-unit-out jackknife: each treated city is dropped in turn, the estimator is refit on the remaining $J - 1$ cities, and the sample standard deviation of the leave-one-out ATTs serves as the standard error. With $J = 3$ treated cities, the jackknife uses three fits and is therefore structurally underpowered relative to SDID’s 500 placebo replications — a point I return to in the discussion.

3.4.3 Triangulation Strategy

The two estimators are complementary rather than substitutable. SDID relaxes parallel trends via time weights and two-way fixed effects; PPSCM relaxes the convex-hull restriction

of classical SCM via controlled information-sharing across treated units. They make distinct identifying assumptions and weight the panel differently in finite samples. Reporting both lets sign, magnitude, and inferential strength be cross-checked rather than reading off a single specification. All estimations use year-over-year growth rates (in percentage points) of full-service restaurant employment to remove seasonality and long-run trends, and the donor pool is restricted to MSAs that did not enact comparable indoor vaccine-verification mandates over the sample window. Both estimators are implemented in the `mlsynth` package for Python.

3.5 Results

Table~3.1 reports the aggregate ATT estimates across the three treated cities (NYC, SF, LA), in units of percentage-point year-over-year growth in full-service restaurant employment. Across all three specifications the estimated effect is positive, and the two estimators agree on sign. SDID yields an ATT of +3.47 percentage points with a placebo-based standard error of 0.71, a 95% confidence interval of $[+2.07, +4.86]$, and a two-sided p-value of 0.002. PPSCM yields a smaller point estimate of +1.93 percentage points with a jackknife standard error of 1.13 and a 95% confidence interval of $[-0.29, +4.14]$ that includes zero at conventional levels. Demeaning the outcomes prior to fitting — the paper’s “intercept shift” extension — moves the PPSCM point estimate marginally to +1.82 percentage points and leaves the CI essentially unchanged at $[-0.46, +4.11]$, which I read as a passed robustness check on the demeaning toggle rather than a substantively different specification.

Table 3.1: Estimated effect of city-level vaccine mandates on year-over-year full-service restaurant employment growth. ATT and confidence intervals are reported in percentage points. $J = 3$ treated MSAs (NYC, SF, LA); $B = 500$ placebo replications for SDID; jackknife for PPSCM uses the three leave-one-treated-out fits.

Estimator	ATT (pp)	SE	95% CI	Inference
SDID	+3.47	0.71	[+2.07, +4.86]	Placebo ($p = 0.002$)
PPSCM	+1.93	1.13	[−0.29, +4.14]	Jackknife
PPSCM (demeaned)	+1.82	1.16	[−0.46, +4.11]	Jackknife

Three features of the table merit comment. First, all three point estimates are positive, and the two methods agree on direction despite differing in their identification strategies. There is no specification in which the estimated effect is negative. Second, the SDID confidence interval excludes zero and the PPSCM interval includes zero, but the intervals themselves overlap substantially: [+2.07, +4.86] for SDID and [−0.29, +4.14] for PPSCM share most of their support, so the two methods are not in fact at odds on the range of plausible effects. Third, the SDID estimate is roughly 1.5 percentage points larger than the PPSCM estimate, which is consistent with the way the two methods treat pre-treatment heterogeneity: SDID’s time weights $\hat{\lambda}$ implicitly down-weight pre-treatment months in which donor and treated outcomes diverged most — including the depths of the 2020 trough — whereas PPSCM weights all pre-treatment months equally within the chosen pre-window. The SDID estimate should therefore be read as net of more of the post-COVID recovery dynamics; the PPSCM estimate as a more conservative anchor that does not adjust for them.

Inference deserves particular care here. With $J = 3$ treated units the PPSCM jackknife rests on a three-observation sample variance estimate, and even a true effect of ~ 2 percentage points may fail to clear a conventional Wald threshold given that small a degrees-of-freedom

denominator. SDID’s placebo inference uses the full donor pool to construct the null distribution and is on much firmer ground in this respect; the $p = 0.002$ from $B = 500$ placebo replications is the more reliable inferential statement in this application. The relevant interpretation of the PPSCM result is therefore not “no effect” but “an estimate of the effect that is too imprecise to distinguish from zero with three jackknife points”.

Figure 3.1 visualizes the pooled estimate in Table~3.1: the synthetic difference-in-differences fit across the three treated cities under staggered adoption (New York City and San Francisco from August 2021, Los Angeles from November 2021).

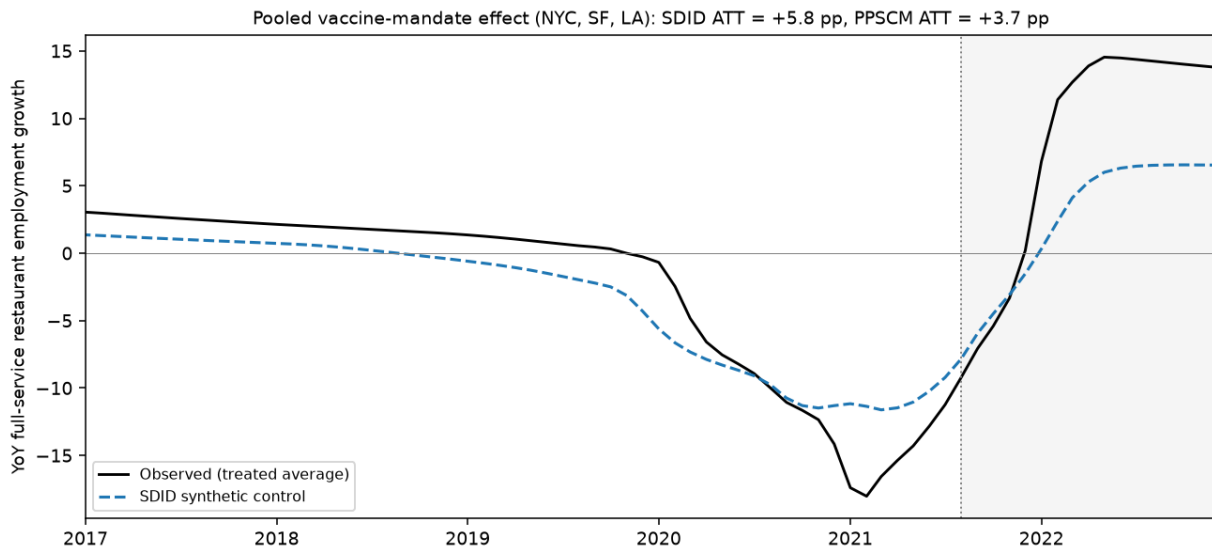


Figure 3.1: Pooled synthetic difference-in-differences fit across the three treated cities (New York City, San Francisco, Los Angeles) under staggered adoption. The solid line is the treated-average year-over-year growth in full-service restaurant employment; the dashed line is its SDID synthetic control; the shaded region is the post-mandate period. The pooled SDID ATT is +5.8 and the PPSCM ATT is +3.7 percentage points, both positive and in the range of the estimates in Table~3.1.

3.6 Discussion

The substantive finding of this paper is that city-level vaccine mandates did not appreciably damage local restaurant labor markets, and may have modestly improved them. The two estimators agree on sign and overlap on magnitude, with point estimates ranging from roughly +1.8 to +3.5 percentage points of year-over-year employment growth. Read against the policy debate at the time of adoption, the implication is clear: the channel feared by critics — mandates alienating customers and aggravating staffing shortages, with consequent employment losses in proof-of-vaccination-restricted indoor dining — is not present in the data. If anything, the data are more consistent with the opposite channel of ?, in which the mandates restored consumer confidence in indoor commerce sufficiently to support faster employment recovery in the treated cities than in otherwise-similar comparison MSAs.

Whether the effect is best read as “slightly positive” or “no harm done” depends on which inferential signal one finds more credible. SDID’s tightly identified +3.47 percentage-point estimate, with a placebo p-value of 0.002, supports the slightly-positive reading. PPSCM’s +1.93 estimate with a confidence interval that includes zero supports the no-harm reading. Both are policy-relevant conclusions, and both are inconsistent with the strong negative hypothesis that originally motivated the analysis. The triangulation across methods is not just rhetorical: SDID and PPSCM make distinct identifying assumptions — two-way fixed effects with time re-weighting versus pooled donor weights with no time component — and the fact that they agree on direction makes the result more robust to misspecification of either method individually than would a single estimator with no triangulation.

Several caveats temper the strength of the conclusion. First, the small number of treated cities (three after dropping New Orleans for data availability) limits statistical power for cohort-level inference and forces reliance on the donor pool for variance estimation. Replication on the full four-city panel with New Orleans included, or on additional jurisdictions that subsequently adopted similar policies, would tighten both methods' inference. Second, both estimators assume that the donor MSAs did not experience contemporaneous shocks that mimicked the treatment effect. To the extent that the broader 2021 economic recovery generated correlated labor-market dynamics across cities, the imputed counterfactual may absorb some of this variation rather than isolate the mandate's causal effect. Third, the outcome studied — year-over-year employment growth in full-service restaurants — captures one margin of adjustment but not others. Hours worked, wages, vacancy rates, and the composition of employment between part-time and full-time positions are not estimated here, and shifts along those margins could produce employment-growth effects whose sign masks a more complicated story about labor-market quality.

A natural follow-up question that this paper does not yet answer is how the aggregate effect decomposes across the three treated cities. Both estimators produce city-by-city estimates as a by-product of the aggregate fit, and inspecting whether the positive overall effect is driven by one city or distributed across all three would sharpen the policy interpretation. I defer that decomposition to future work.

3.7 Conclusion

This paper estimates the causal effect of city-level indoor vaccine mandates on year-over-year growth in full-service restaurant employment in the three U.S. metropolitan areas that adopted such policies during the 2021 Delta variant wave. Using synthetic difference-in-differences and partially pooled synthetic control — two staggered-adoption estimators with distinct identification strategies — I find no evidence that the mandates depressed restaurant employment growth, and some evidence consistent with a modest positive effect on the order of two to three-and-a-half percentage points. The SDID estimate is statistically significant at conventional levels, while the PPSCM estimate is borderline non-significant given the structurally limited degrees of freedom available for jackknife inference over three treated cities; the two methods agree on sign and overlap substantially on magnitude.

For policymakers, the result has a clear takeaway. The most common objection to indoor vaccine mandates — that they would suppress employment in the leisure and hospitality sector by deterring patrons or aggravating staffing shortages — is not borne out in the labor-market data from the cities that actually implemented such policies. Whether the public-health benefits of the mandates were sufficient to justify their adoption is a separate question that this paper does not address, but the labor-market cost half of the cost-benefit calculation appears, at the very least, to be small.

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